FedECS-LT: Exact Class Statistics for Long-Tailed Federated Learning with Heterogeneous Client Models

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Abstract

With different client architectures and long-tailed, skewed labels, federated learning loses most of its rare-class accuracy. We study generator-based model-heterogeneous federated learning, where clients share a frozen extractor and a federated conditional feature generator, and each architecture trains a head. We locate three candidate causes: Adam with coupled weight decay shrinks the generator's class-specific parameters where a class is absent; each head's last layer fits one client's label prior; and generated features that do not carry their label penalise the spread of the logits. We propose FedECS-LT, which replaces failing estimates by class statistics that secure aggregation sums exactly: a generator anchored to the class means, moment correction of generated features, and, as its last step, a class-balanced linear readout of every architecture's last layer. Its heads add logit adjustment, consensus soft labels and ten synthetic epochs per round. With ten architectures, one client each, FedECS-LT reaches 52.6 ± 1.3 % balanced accuracy on CIFAR-10-LT and 18.7 ± 0.2 % on CIFAR-100-LT, +15.1 and +8.6 points above the strongest of eleven other methods without the readout (Holm-corrected $p = 0.0052$ and 0.0042). Most of this margin is the readout's: with the same readout added to every method of the GeFL-F pipeline, FedECS-LT leads GeFL-F by +2.3 and +3.8 points ($p = 0.34$ and 0.0095) and the strongest method by +1.4 and +1.5, unresolved with three seeds, at 185 times the communication of locally logit-adjusted heads. On six naturally imbalanced datasets the same-readout margins are -0.03 to +0.9 points. With one shared model, 6 other methods are better on CIFAR-10-LT.

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1. Introduction

Federated learning (FL) trains a model on data that stay with their owners [1, 2]. Two properties of real deployments make it hard. The participants run different networks on different hardware, so their parameters cannot be averaged. Their label distributions differ and follow a long tail, so the rarest classes are held by few participants and in small numbers. Each property has its own line of work: model-heterogeneous FL [3, 4, 5, 6, 7, 8] lets clients with different architectures share knowledge through prototypes, logits, generators or shared sub-networks, and federated long-tailed learning [9, 10, 11, 12, 13] corrects the bias that a skewed, long-tailed label distribution induces in a single shared model. The two are rarely studied together: model-heterogeneous methods are evaluated without a global long tail [14], and federated long-tail methods assume one shared model.

We study generator-based model-heterogeneous FL, in the form of GeFL-F [8]. Its clients first train a small feature extractor by federated averaging (FedAvg) and freeze it; they then train a conditional feature generator by FedAvg; finally every client trains a head of its own architecture on generated features and on its own real features. The generator is the only channel through which an architecture learns about classes its clients lack. GeFL-F was evaluated on balanced clients. Under long-tailed label skew we find that it loses most of its accuracy on the rare classes, and we ask where.

Where the rare classes are lost. Because a client partition is a pure label shift, the target of every class-conditional quantity does not depend on the partition (Section 3); its federated estimate can. We examine how it can still lose rare classes: through the optimiser, through parameters tied to a client's label prior, through the synthetic data and through sample size. We locate three such losses, one through each of the first three routes (Section 4).

1. *The generator.* With Adam [15] and coupled weight decay, a client that lacks a class moves the generator's class row, its only class-specific parameter, toward zero by about one learning rate per step, and flat averaging includes these clients. In GeFL-F's runs the rare classes' rows end several times shorter than the frequent ones'; whether this causes their loss is not yet tested.

2. *The last layer.* Each head is trained on one client's real data, so its last layer fits that client's label prior. Per-client logit corrections (logit adjustment [16], FedLC [17], Balanced Softmax [18] and, up to a step size, FedRS [19]) change only the initialisation of this layer and bring in no information about the classes the client lacks.
3. *The synthetic data.* When generated features do not carry their label, the synthetic phase of head training penalises the spread of the logits instead of teaching the missing classes.

Our method. FedECS-LT (Exact Class Statistics for Long-Tailed FL) replaces each estimate that fails by a class statistic that secure aggregation [20, 21] computes exactly: the per-class counts, feature sums and sums of squared feature norms over all clients. The generator is anchored to the class means and has no class row for the decay to shrink; when the classes are poorly separated, a gate computed from the same statistics falls back to class rows averaged over their holders only (class-count weighted averaging, CCWA). Generated features are rescaled to the exact class means and spreads, and heads are trained with logit adjustment and with consensus soft labels, the averaged prediction of all architectures. As its last step, FedECS-LT sets every architecture's last layer to the exact readout, a class-balanced linear discriminant computed from aggregated sums of its embeddings. None of these statistics is new on its own: aggregated class statistics underlie classifier calibration [22], closed-form federated classifiers [23, 24, 25], including an LDA-type classifier from class statistics [25], and prototype methods [3], and statistics-conditioned generators have been proposed for heterogeneous clients [26]. What we add is the diagnosis of *which* parameters fail under a long tail and the use of exact statistics at exactly those places, including a readout per architecture that needs statistics pooled across architectures.

Evaluation. We re-implemented the pipeline and every comparison of Section 7 in the Flower framework [27], with the published methods taken from their authors' or library code (HtFLlib [14], FedLTLib [28], whose code is LongTailLib). We evaluate in two protocols: the model-heterogeneous setting of GeFL-F (ten client architectures; long-tailed CIFAR-10 and CIFAR-100 and, with natural class imbalance, DermaMNIST, OrganAMNIST, BloodMNIST, PathMNIST, EMNIST (ByClass) and FER-2013; eleven other methods) and the standard federated long-tailed protocol with one shared ResNet-8 (long-tailed CIFAR-10 and CIFAR-100 at imbalance ratio 100; eleven other

methods). Every result is a mean over seeds 3–5 (fewer where a table says so), with paired tests and a Holm correction, and every metric is logged per class.

Contributions.

1. A long-tail evaluation of generator-based model-heterogeneous FL and a measured, stage-by-stage diagnosis of where it loses rare classes, supported by elementary lemmas (Section 4).
2. FedECS-LT, which uses exact, securely aggregated class statistics in the generator and the readout, and trains the heads with logit adjustment, consensus soft labels and more synthetic epochs (Section 5).
3. A comparison against twenty-one published methods in two protocols: eleven on eight datasets with different client models, and eleven on two benchmarks with one shared model (in the authors' LongTailLib code), each protocol on the same client splits, with paired statistics, per-class metrics and the cost of every method (Section 7). Every method of the GeFL-F pipeline is also evaluated with the readout of FedECS-LT added, which separates the readout's gain from that of the rest of the method. With different client models, FedECS-LT has the highest mean balanced accuracy on both long-tailed benchmarks and all six naturally imbalanced datasets, and most of its margin comes from the readout: with the same readout for every method of the GeFL-F pipeline, its lead of +1.4 to +1.5 points over the strongest other method on the long-tailed benchmarks is not significant with three seeds. With one shared model, 6 other methods are better on CIFAR-10-LT (FedYoYo, FedLoGe, CCVR, FedETF, FedAvg and MOON).

The code, every log and every figure's data are released (see the data and code availability statement).

2. Related work

Model-heterogeneous federated learning. When clients run different architectures, their parameters cannot be averaged and knowledge is exchanged in another form. Distillation methods exchange predictions on public or synthetic inputs (FedMD [29], FedDF [30]). Prototype methods exchange class means of embeddings (FedProto [3]) or learn global prototypes on the server (FedTGP [4]). Other methods share a part of the network: the head (LG-FedAvg [31], FedGH [6]) or a small homogeneous model next to each client's

own (FedMRL [7]). Generator-based methods train a generator on the server against the ensemble of client predictors (FedGen [5]; FedFTG [32] does so for one shared model, in input space), let each client train a decoder whose latent prior is centred on server-averaged class prototypes (FedVTC [26], which generates images for generalisation to unseen data rather than for long-tailed classes), or train a conditional generator on the clients by FedAvg (GeFL, GeFL-F [8]). HtFLlib [14] benchmarks six of these methods under label skew, with personalised evaluation on each client's test split and the best round, but not under a long-tailed global distribution. FedECS-LT keeps GeFL-F's three-stage design and changes how its class-specific quantities are estimated.

Federated learning under label skew and long tails. With one shared model, the classifier is the most biased part under label skew. Classifier calibration re-trains it on virtual features drawn from aggregated class statistics (CCVR [22]) or on federated features learned on the server (CReFF [9]). Closed-form classifiers are computed from aggregated sums of pre-trained features (Fed3R [23], a ridge-regression classifier; AFL [33], a least-squares head; FedNCM [24], nearest class means; FedCOF [25], an LDA-type classifier with a shrunk within-class covariance estimated from class means). These works also note that such sums do not depend on the partition. Our readout is the shared-covariance discriminant of the same family, with equal class priors, computed for each architecture on trained rather than pre-trained bodies, so every client runs every architecture's body on its own features. FedETF fixes the classifier to a simplex equiangular tight frame [34]. Loss corrections use each client's label frequencies (FedLC [17], FedRoD [35]) or the set of classes it holds (FedRS [19]), and long-tail methods re-balance gradients (Fed-GraB [10]), decouple a generic and a local head (FedLoGe [11]), combine prototype alignment with classifier rectification (RUCR [13]) or distil between augmented views of local data (FedYoYo [12]). FedLTLib [28] benchmarks most of them in one protocol, which we follow for the one-shared-model setting. Sampling features from Gaussians with the pooled class moments, as in CCVR and DFL-FS [36], is the closest prior to our moment correction, which instead maps a learned generator's samples onto the same moments. Class-specific weights are also known to degrade when clients lack classes (FedAwS [37], FedRS).

Long-tailed learning, optimisers and synthetic data. FedECS-LT builds on logit adjustment [16], effective numbers of samples [38] and the decoupling

Table 1: Methods in this study. Different models: clients may run different architectures. Long tail: designed for long-tailed or skewed label distributions. Synthetic: trains or uses synthetic data or features. Extra data: needs data that no client holds (a public or proxy set). [a]FedDF needs public data; here it distils on GeFL-F's generated features.

Method	Different models	Long tail	Synthetic	Extra data
FedAvg [1], FedProx [45], MOON [46]	–	–	–	–
FedLC [17], FedRS [19], Balanced Softmax [18]	–	✓	–	–
CCVR [22], CReFF [9]	–	✓	✓	–
FedETF [34], Fed-GraB [10], FedLoGe [11]	–	✓	–	–
RUCR [13], FedYoYo [12]	–	✓	–	–
FedProto [3], FedTGP [4]	✓	–	–	–
LG-FedAvg [31], FedGH [6], FedMRL [7]	✓	–	–	–
FedGen [5], GeFL-F [8]	✓	–	✓	–
FedDF [30]	✓	–	✓[a]	✓[a]
FedECS-LT (ours)	✓	✓	✓	–

of representation and classifier [39]. The generator analysis rests on the difference between coupled L2 regularisation and decoupled weight decay in Adam [40, 15]. Class-conditional generators can collapse when trained on little data, even centrally [41], and classifiers trained on generated data show how much class information the samples carry [42]. Weighting clients by their class counts when combining class-specific knowledge goes back to FedGen's ensemble weights [5]; the gate statistic is a trace form of Fisher's discriminant ratio [43]. Generators matched to kernel mean embeddings of the data have been used for private data release [44].

Table 1 places the methods we compare against.

3. Problem setting

Clients and data. There are K clients and C classes. Client k holds a dataset D_k with n_{kc} samples of class c; $n_k = \sum_c n_{kc}$, $n_c = \sum_k n_{kc}$ and $N = \sum_c n_c$. The global class sizes follow a long tail, $n_c \propto \rho^{-c/(C-1)}$ with imbalance ratio $\rho = n_0/n_{C-1}$, and each class is divided among the clients by a Dirichlet draw with concentration α [47]. $H_c = \{k : n_{kc} > 0\}$ is the set of the $m_c = |H_c|$ clients that hold class c. Client k trains a model of architecture $a(k)$, one of G architectures. The goal is a model per architecture with high *balanced* accuracy, the mean recall over classes, on a test set in which every class is equally important.

The pipeline of GeFL-F. GeFL-F [8] trains in three stages.

1. *Stage I.* A small feature extractor ϕ is trained together with the heads by FedAvg for T_{I} rounds and then frozen. Every client maps its images to features $h = \phi(x)$.
2. *Stage II.* A conditional variational autoencoder [48] $G(h \mid y)$ is trained on the features by FedAvg for T_{II} rounds, with Adam and coupled weight decay. The label enters through a linear projection; its weights for class c, which we call the *class row* r_c, are the only class-specific parameter.
3. *Stage III.* In each of T_{III} rounds every client trains its head for T_s epochs on generated features with uniformly drawn labels and then for T_r epochs on its real features; heads of the same architecture are averaged. Each head consists of a *body* of convolutional blocks, which maps h to an embedding e, and a linear last layer.

A client split is a pure label shift. Within a class, the split assigns samples to clients uniformly at random. So for every client k and class c with $n_{kc} > 0$, the client's class-c samples are a uniformly random subset of all class-c samples, and $P_k(x \mid y) = P(x \mid y)$; the same holds for the features $h = \phi(x)$, since ϕ is shared. A class-conditional quantity, such as a class mean, a class row or a readout, therefore has the same target on every client as on the pooled data. Federation can make such a quantity noisier or slower to learn. We examine four ways in which it can also make it *biased*: (i) the optimiser, (ii) parameters coupled to a client's label prior, such as biases and last layers, (iii) what the synthetic data teach, and (iv) sample size, which biases nonlinear estimates such as a spread or an inverse covariance in small classes. Section 4 locates the first three in GeFL-F; we do not claim that the list is complete (the drift of the shared extractor in Stage I, for example, is not covered).

Secure aggregation. Secure aggregation [20, 21] lets the server learn the sum of the clients' vectors and nothing else about any single vector. A statistic that is a sum over clients of per-class quantities is therefore computed exactly, and equals its value on the pooled data, whatever the partition. FedECS-LT uses only such statistics.

What every client must run. FedECS-LT lets clients use the heads of other architectures: each client receives every architecture's current head in Stage III (for the consensus soft labels) and runs every architecture's body once on its own features (for the readout). We therefore assume that every client

can store and evaluate every head, though it trains only its own, and that the architectures are not secret. Clients whose hardware cannot evaluate the largest head are outside this setting. In the main setting each architecture has one client ($K = G = 10$), so heads are never averaged within an architecture there.

4. Where generator-based heterogeneous FL loses rare classes

This section locates the three losses that FedECS-LT addresses. Each comes with a short elementary lemma (proofs in Appendix A) and a measurement on the GeFL-F runs of Section 7.

4.1. The generator: coupled decay shrinks rare classes' class rows

In one round, client k starts from the global row r_c, runs local Adam steps and returns $r_{k,c}$. Let $\Delta_k = \|r_{k,c}\| - \|r_c\|$, let $\bar{G}_c$ be the mean of Δ_k over the holders H_c and $\bar{D}_c$ the mean of $-\Delta_k$ over the other clients, and $p_c = m_c/K$.

Lemma 1 (contraction under flat averaging)**.** *Under* $r_c^+ = \frac{1}{K}\sum_k r_{k,c}$, $\|r_c^+\| \le \|r_c\| + p_c\bar{G}_c - (1-p_c)\bar{D}_c$, *with equality if and only if all* $r_{k,c}$ *are non-negative multiples of one another. The row contracts whenever* $p_c < \bar{D}_c/(\bar{G}_c + \bar{D}_c)$.

The bound only accounts for the measured norm changes; how large $\bar{D}_c$ is depends on the optimiser.

Lemma 2 (coupled decay under Adam)**.** *On a client without class* c*, the gradient of the loss with coupled L2 regularisation* λ *with respect to* r_c *is exactly* λr_c*. Since Adam is invariant to a positive rescaling of the gradient sequence up to its* ϵ*, each step of such a client moves every coordinate of* r_c *toward zero by about the learning rate* η*, independently of* λ*, as long as* $\lambda|r_{c,j}| \gg \epsilon$*. With* $\lambda = 0$ *this shrinkage vanishes; with decoupled weight decay [40] it becomes a factor* $1 - \eta\lambda$ *per step, which is negligible here.*

A holder's data gradient fits the row and need not increase its norm, so $\bar{G}_c$ can be much smaller than $\bar{D}_c$, and rare classes' rows can contract even when most clients hold the class; a threshold on m_c alone cannot decide it. Counting Adam steps gives a predictor that uses only the class counts and the number of local steps q_k: with f_{kc} the probability that a batch of client k contains class c,

$$\kappa_c = \frac{\sum_{k \in H_c} q_k f_{kc}}{\sum_{k \notin H_c} q_k}, \tag{1}$$

and the row drifts toward zero in every coordinate when $\kappa_c < 1$. This count of Adam steps is a heuristic, not a bound, and we have not checked it against the measured rows: it ignores the moments Adam carries between rounds (Appendix A). Centrally the same decay is harmless, because Adam's second-moment estimate for r_c is dominated by data gradients as long as batches that contain the class occur within the memory of that estimate, about $1/(1-\beta_2)$ steps for Adam's second-moment parameter β_2.

Figure 1 shows the effect on GeFL-F: after Stage II the class rows of the rarest classes are several times shorter than those of the most frequent classes. Shorter rows are not yet a cause of the rare classes' loss. A classifier trained by the server on real validation features (the referee, Appendix C.1) recognises GeFL-F's generated CIFAR-10-LT features at 12.9 $\pm$ 2.4 on average, with 9.4 $\pm$ 0.9 on the head classes and 16.8 $\pm$ 0.3 on the tail classes, and at 1.1 $\pm$ 0.1 on CIFAR-100-LT, about chance: the generator carries little class information for any class, not only for the rare ones. The row shrinkage is therefore one measured property of GeFL-F's generator, not a demonstrated cause of its rare-class loss; an arm with decoupled decay alone would test it directly. Two remedies follow. Decoupling the decay removes the shrinkage. Averaging each row over its holders only, with weights that grow with the holder's count (class-count weighted averaging, CCWA), removes the non-holders' term $\bar{D}_c$ from Lemma 1 and keeps coupled decay:

$$r_c \leftarrow \frac{\sum_k E(n_{kc})\, r_{k,c}}{\sum_k E(n_{kc})}, \qquad E(n) = \frac{1-\beta^n}{1-\beta}, \quad \beta = 0.999, \tag{2}$$

where $E(n)$ is the effective number of samples [38]: zero for a client without the class, about n for small n, and bounded by $1/(1-\beta)$ so that one large client cannot dominate a class. Both sums in Eq. (2) are computed by secure aggregation.

4.2. The last layer: logit corrections are reparametrisations

Lemma 3 (logit corrections are reparametrisations)**.** *Let a head have a free last-layer bias b, trained by SGD without decay on b, and let $a_k \in \mathbb{R}^C$ be a constant of client k, as in logit adjustment ($a_k = \tau \log \pi_k$) [16], FedLC [17] or Balanced Softmax [18]. Training with the loss $\ell(f + a_k, y)$ from (θ_0, b_0) produces exactly the trajectory of training with $\ell(f, y)$ from $(\theta_0, b_0 + a_k)$: every parameter except b is the same at every step. FedRS, which scales the logits of absent classes by $\alpha \in (0, 1]$ [19], is cross-entropy in the rescaled last-layer*

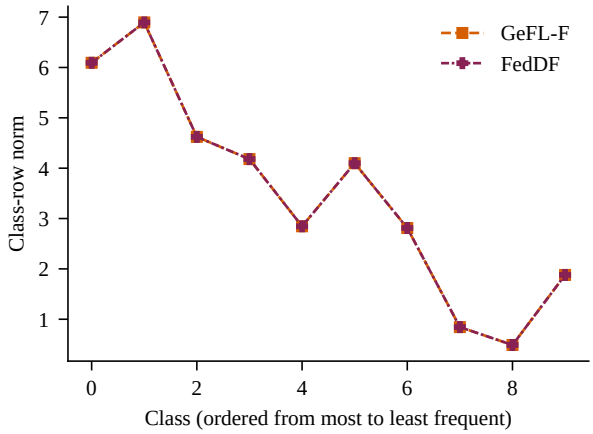


Figure 1: L2 norm of each class's class row of GeFL-F's generator after Stage II, classes ordered from the most to the least frequent (CIFAR-10-LT, imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$; mean over seeds 3–5). FedDF uses the same generator. FedECS-LT's anchored generator has no class rows.

variables $\alpha w_c, \alpha b_c$ of those classes, trained with the smaller learning rate $\alpha^2\eta$: it changes the initialisation and the step size of the absent classes' rows, not what the client knows about them.

For a head trained on one client's real data, a logit shift (logit adjustment, FedLC, Balanced Softmax) is therefore equivalent to plain training from the shifted bias $b_0 + a_k$ followed by subtracting a_k from the logits at test time, which is post-hoc logit adjustment [16] of a head trained from a shifted initial bias. The body can still learn a different representation from the one that plain training from the same bias would learn, because the shifted logits change the gradients that reach it, but the correction brings in no information about the classes the client lacks. Such a head is an extreme case of minority collapse [49]: for the classes the client lacks, the imbalance ratio is infinite. The last layer can instead be computed from aggregated sums.

Lemma 4 (partition invariance)**.** *An estimator that is a function of a sum $S = \sum_k \sum_{i \in D_k} T(x_i, y_i)$, with T fixed in advance, takes the same value for every partition of the pooled data, and equals the centralised estimator.*

This property is known for closed-form federated classifiers [23, 33, 25]; we state it because the method relies on it in several places.

The exact readout of FedECS-LT (Section 5.4) is such an estimator, and so are Fed3R [23], FedNCM [24], FedCOF [25] and, for a fixed sampling

seed, CCVR's virtual features [22] when computed over all clients. With it, every architecture's last layer equals the same readout computed centrally on the same body, so differences in accuracy with this readout come from the bodies. The readout alone raises GeFL-F's balanced accuracy by +16.6 points on CIFAR-10-LT and +7.2 on CIFAR-100-LT (Table 8).

4.3. The synthetic data: label-free features penalise the logits' spread

Lemma 5 (label-free synthetic data)**.** *If the synthetic label y is uniform on $\{1, \dots, C\}$ and independent of the generated feature, then for every logit vector f, $\mathbb{E}_y\, \ell(f, y) = \mathrm{LSE}(f) - \bar{f} = \log C + \mathrm{KL}(u \,\|\, \mathrm{softmax} f) = \log C + \frac{1}{2C} \sum_c (f_c - \bar{f})^2 + O(\|f - \bar{f}\|^3)$, where $\mathrm{LSE}(f) = \log \sum_c e^{f_c}$, $\bar{f}$ is the mean logit and u the uniform distribution.*

When generated features carry little information about their label, the synthetic phase therefore pulls the logits together, that is, it reduces the *logit spread* (the mean over images of the standard deviation of the logits over classes), instead of teaching the classes a client lacks. GeFL-F's generated features are of this kind (the fidelity above). Alternating the synthetic phase with the real phase decides what survives. In an illustrative model with quadratic losses (the heads are nonlinear and trained with cross-entropy), the round map converges to a point where the synthetic data alone decide the directions that the client's real loss leaves free (Appendix A). For a head trained on one client's real data, the directions of the absent classes are nearly free in this sense, but not exactly: cross-entropy still pushes their logits down. Logit adjustment shifts the real phase's targets for the absent classes; without it, cross-entropy on data that a head separates keeps pushing those logits down [50]. Figure 2 shows the first effect on the real pipeline: in every round on CIFAR-10-LT the synthetic phase reduces the logit spread of the heads on the clients' real features, and it reduces it less for FedECS-LT than for GeFL-F. FedECS-LT also differs from GeFL-F in the number of synthetic epochs, the consensus soft labels and logit adjustment, so this comparison does not isolate the effect of features that carry their label. FedDF, which trains on GeFL-F's generator, loses less spread than FedECS-LT: the change of the spread is therefore not determined by whether the generated features carry their label.

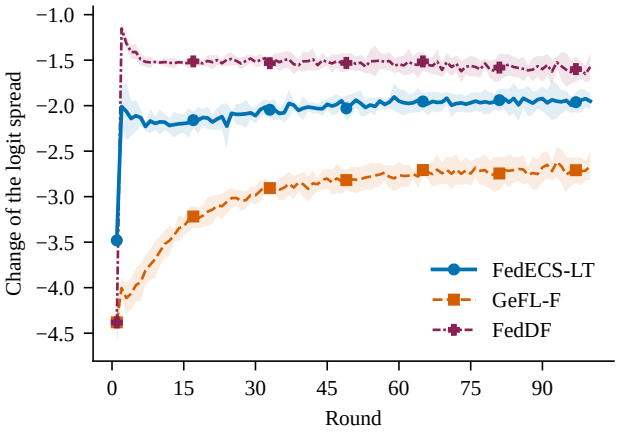


Figure 2: Change of the logit spread (the mean over images of the standard deviation of the logits over classes) of the clients' heads on their own real features during the synthetic phase of each Stage-III round (after minus before; negative: the synthetic phase pulled the logits together), per round (CIFAR-10-LT, imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$; mean over clients and seeds 3–5, band: $\pm$ one standard deviation).

5. FedECS-LT

FedECS-LT keeps the three stages of GeFL-F and the frozen extractor, adds a closing step that sets every last layer, and changes how the class-specific quantities of the generator and the last layer are obtained: it uses statistics that secure aggregation computes exactly instead of parameters estimated by FedAvg. The heads are trained with logit adjustment (each client's own prior), consensus soft labels and ten synthetic epochs per round. Algorithm 1 lists the pipeline and Fig. 3 shows it.

5.1. Statistics release

After Stage I, client k computes, for every class c, its count n_{kc}, the sum of its features $S_{kc}^{(1)} = \sum_{i \in D_{kc}} h_i$ and the sum of their squared norms $S_{kc}^{(2)} = \sum_{i \in D_{kc}} \|h_i\|^2$. Secure aggregation returns the totals n_c, $S_c^{(1)}$ and $S_c^{(2)}$, from which the server obtains the exact class means and spreads

$$\mu_c = S_c^{(1)}/n_c, \qquad V_c = S_c^{(2)}/n_c - \|\mu_c\|^2 = \mathbb{E}\Big[\|h - \mu_c\|^2 \,\Big|\, y = c\Big]. \tag{3}$$

These equal the values on the pooled data (Lemma 4) up to the quantisation of secure aggregation: each client's contribution is divided by a public bound and quantised to 30 bits with unbiased stochastic rounding, an error of at

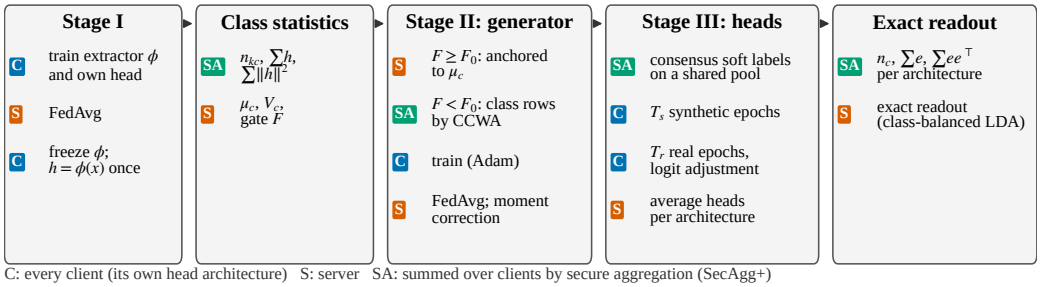


Figure 3: Overview of FedECS-LT. C: computed by each client; S: computed by the server; SA: summed over the clients by secure aggregation. The class statistics released after Stage I give the class means and spreads and the gate F, which chooses the generator of Stage II. The moment correction is computed once after Stage II. In Stage III the heads train on corrected generated features with consensus soft labels and on real features with logit adjustment. Last, the exact readout sets every architecture's last layer from aggregated sums of its embeddings.

most 1.5×10^{-8} of the bound per client and coordinate. Values beyond the bound would be clipped, and with fewer than three participants the sum is computed in clear, since secure aggregation needs at least three. We call the statistics exact in this sense. Each client uploads them once, $C(d+2)$ numbers for features of dimension d.

5.2. Generator

Separation gate. A generator anchored to the class means separates two classes by at most a constant times the distance between their means (Lemma 7 in Appendix A). When the means are close relative to the within-class spread, the anchor alone carries little class information. The server therefore computes the class separation

$$F = \frac{\frac{1}{C}\sum_c \|\mu_c - \bar{\mu}\|^2}{\frac{1}{C}\sum_c V_c}, \qquad \bar{\mu} = \frac{1}{C}\sum_c \mu_c, \tag{4}$$

from the released statistics. If $F \geq F_0 = 0.05$ (fixed before the reported runs and not swept; Section 6.9) it uses the mean-anchored generator; otherwise it uses GeFL-F's generator with class rows averaged by CCWA (Eq. (2)). On our datasets F ranges from 0.014 to 0.449; the gate chose class rows on CIFAR-10-LT with one shared model and FER-2013.

Mean anchor. The anchored generator models only the within-class residual around the exact mean,

$$\tilde{h} = \mathrm{ReLU}\Big(\mu_y + \mathrm{dec}(z,\, A\mu_y)\Big), \qquad z \sim \mathcal{N}(0, I), \tag{5}$$

where the class enters only through its mean and a projection A shared by all classes. The generator has no class-specific parameter, so Lemma 1 has nothing to shrink. It is trained by FedAvg with Adam, as GeFL-F's generator.

Moment correction. After Stage II the server draws 500 samples per class from the generator, measures their mean $\tilde{\mu}_c$ and spread $\tilde{V}_c$, and corrects every later sample of class c to the exact moments,

$$\tilde{h} \leftarrow \mathrm{ReLU}\Big(\mu_c + s_c(\tilde{h} - \tilde{\mu}_c)\Big), \qquad s_c = \mathrm{clip}\Big(\sqrt{V_c/\tilde{V}_c},\, 0.1,\, 10\Big). \tag{6}$$

Without the final ReLU and the clipping this is the map of least displacement that attains the exact mean and spread (Lemma 8 in Appendix A); with them, the corrected samples match these moments only approximately: after the correction the generated within-class spread is 0.73 to 0.98 times the exact one (median 0.86) over every class and run of the long-tailed benchmarks. The server sends $(\mu_c, \tilde{\mu}_c, s_c)$ to the clients, which hold the generator and draw corrected samples themselves.

5.3. Heads

Each round, client k first takes $T_s\lfloor n_k/B \rfloor$ SGD steps on batches of B corrected generated features ($B = 128$ on CIFAR, 64 on the other datasets) ($T_s = 10$ synthetic epochs of the size of its real data; GeFL-F uses $T_s = 1$) and then trains $T_r = 5$ epochs on its real features, each phase with a fresh SGD optimiser. The real phase uses logit adjustment with the client's own prior [16], $\ell(f + \tau \log \hat{\pi}_k, y)$ with $\tau = 1$ and the add-one smoothed class frequencies $\hat{\pi}_k$ of the client. The synthetic phase, whose labels are drawn uniformly over the classes, uses cross-entropy without adjustment, and its target is a *consensus soft label*

$$t(\tilde{h}) = (1-\omega)\, e_y + \omega\, \frac{1}{G} \sum_{g=1}^{G} \mathrm{softmax}\Big(f_g(\tilde{h})\Big), \qquad \omega = 0.5, \tag{7}$$

where e_y is the one-hot generated label and f_g the current head of architecture g: the federation's averaged prediction on that feature, the target

form of AvgKD [51] and FedDF's ensemble distillation [30], here computed on generated features through one secure sum. Every client regenerates the same pool of generated features from a seed sent by the server and returns $w_k \operatorname{softmax}(f_{a(k)}(\tilde{h}))$ with $w_k = 1/(G\,|\{j : a(j) = a(k)\}|)$; the secure sum over clients is the average in Eq. (7). Every client thus receives every architecture's current head, which FedAvg within an architecture does not reveal (Section 3); secure aggregation only hides each client's predictions from the server. Heads of the same architecture are averaged as in GeFL-F.

5.4. Exact readout

The last layer is where GeFL-F loses most of the rare classes (Section 4.2), and it is the one part of a head that class statistics determine in closed form. FedECS-LT therefore ends with the *exact readout* (ER), a class-balanced linear discriminant computed from securely aggregated sums. After Stage III, the server broadcasts every architecture's body; each client computes, for every architecture g, the per-class counts, the sums $\sum e$ and the second moment $\sum ee^\top$ of the embeddings $e \in \mathbb{R}^{d_e}$ (the input of the last layer) of its real features, and secure aggregation sums them over all clients. With the class means ν_c and the pooled within-class covariance Σ shrunk to $\Sigma_\gamma = (1-\gamma)\Sigma + \gamma \frac{\operatorname{tr}\Sigma}{d_e} I$, $\gamma = 0.1$, the last layer of architecture g becomes

$$w_c = \Sigma_\gamma^{-1} \nu_c, \qquad b_c = -\tfrac{1}{2}\, \nu_c^\top \Sigma_\gamma^{-1} \nu_c, \tag{8}$$

with $b_c = -10^4$ for a class no client holds. By Lemma 4 it equals the same estimator computed centrally on the pooled embeddings of the same body (not a last layer trained centrally by SGD).

With the population means and covariance and $\gamma = 0$ it would be the Bayes classifier for equally likely classes when the embeddings of each class are Gaussian with a shared covariance. Computed from the sums with shrinkage and a pseudo-inverse, it is a plug-in estimate of that classifier; when the embeddings are not Gaussian with a shared covariance, it is a linear classifier without an optimality guarantee.

The readout needs one extra round and $C(d_e + 1) + d_e^2$ numbers per architecture and client, and every client must run a forward pass (no training) of every architecture's body on its own features once; a client that cannot do so would be left out of the sums. The readout does not use the head's trained last layer, only its body, so it can be added to any method whose heads have a body and a linear last layer. We therefore report FedECS-LT

Algorithm 1 FedECS-LT (one run; all sums are computed by secure aggregation)

Stage I: train extractor ϕ and heads by FedAvg; freeze ϕ; $h = \phi(x)$ on every client
clients release $n_{kc}, S_{kc}^{(1)}, S_{kc}^{(2)}$; server forms μ_c, V_c (Eq. (3)) and F (Eq. (4))
Stage II: if $F \geq F_0$ train the anchored generator (Eq. (5)) by FedAvg **else** train GeFL-F's generator, class rows by CCWA (Eq. (2))
server measures $\tilde{\mu}_c, \tilde{V}_c$ and sends the moment correction $(\mu_c, \tilde{\mu}_c, s_c)$ (Eq. (6))
for round $t = 1, \ldots, T_{\mathrm{III}}$ **do** ▷ Stage III
    server sends heads and a pool seed; clients return their predictions, summed by secure aggregation, for the consensus soft labels (Eq. (7))
    each client: T_s epochs on corrected generated features with consensus soft labels $t(\tilde{h})$, then T_r epochs on real features with logit adjustment
    server averages heads of the same architecture
end for
Exact readout: clients release embedding sums per architecture; server sets every last layer by Eq. (8)

with its readout, its heads before the readout ("heads only"), every method without the readout, and every method of the GeFL-F pipeline with the same readout added (ER), which separates the gain of the readout from that of the rest of the method.

5.5. Cost and privacy

Beyond GeFL-F, FedECS-LT sends the class statistics once, the moment correction once, one vector of C consensus probabilities per generated feature per round, and the readout sums once. Table 12 reports the measured communication and time of every method. The server, and through the generator and the moment correction every client, learns the federation-level class means and spreads. For a class that one client holds, these equal that client's own statistics, and for a class with few images, they reveal more about each image than a mean over thousands would. The features come from a shallow extractor, so a class mean is close to a mean image at reduced resolution; we did not test how much of the images an inversion of the released means recovers. Formal differential privacy for the released statistics is possible with the Gaussian mechanism but is not part of the results

reported here, and it would not cover the FedAvg training of the generator and heads.

6. Study design

The study tests the analysis of Section 4 and the method of Section 5 on the problem of Section 3: clients with different head architectures and long-tailed, skewed label distributions.

6.1. Research questions

All experiments use the same client splits for every method.

- **RQ1 (diagnosis).** Do the three losses of Section 4 occur in GeFL-F's own runs (class-row norms, readout gain and logit spread)?
- **RQ2 (effect).** Does FedECS-LT, which ends with the exact readout, raise balanced accuracy over GeFL-F and the other methods that work with different client models, each without the readout (Section 7.1)?
- **RQ3 (representation or last layer).** How much of the difference remains when every method of the GeFL-F pipeline gets the same exact readout, so that only the heads' bodies differ (Section 7.1, Table 8)?
- **RQ4 (scope).** Does the effect hold with natural class imbalance, and in the standard protocol with one model shared by all clients (Sections 7.3 and 7.4)?

Cost and calibration are reported for every method as secondary outcomes.

6.2. System under evaluation

FedECS-LT keeps the three-stage pipeline of GeFL-F (Section 3), in which the frozen extractor computes every image's features once, without augmentation. It changes five things in this pipeline (items 1–5) and adds a last step (item 6; Section 5):

1. the generator, which is anchored to the exact class means, or uses class rows averaged by CCWA when the gate finds the classes poorly separated;
2. the moment correction of generated features;
3. logit adjustment in the real phase;

Table 2: Settings of FedECS-LT (all datasets; with one shared model, 10 real epochs per round and SGD with momentum 0.9 and the protocol's learning-rate decay). GeFL-F uses $T_s = 1$, cross-entropy and no consensus soft labels; the generator settings are GeFL-F's.

Setting	Value
Separation gate threshold F_0	0.05
CCWA parameter β	0.999
Moment correction: samples per class	500
Moment correction: range of s_c	$[0.1, 10]$
Synthetic / real epochs per round T_s / T_r	10 / 5
Logit adjustment τ	1
Consensus soft-label weight ω	0.5
Readout shrinkage γ	0.1
Generator optimiser, learning rate	Adam, 10^{-3}
Generator weight decay (coupled L2)	10^{-3}
Head optimiser (each phase, each round)	SGD, learning rate 0.1, no momentum

4. consensus soft labels in the synthetic phase;
5. the number of synthetic epochs per round, $T_s = 10$ instead of GeFL-F's 1;
6. the exact readout of Section 5.4, which sets every last layer after Stage III.

Table 2 lists the settings of FedECS-LT, which are the same on every dataset. The extractor and the ten head architectures, CNN-1 to CNN-10 (one to four convolutional blocks of different widths and a linear last layer), follow GeFL-F [8]; Appendix B gives their layers.

6.3. Datasets and client splits

Long-tailed benchmarks. CIFAR-10 [52] and CIFAR-100 [52] are made long-tailed by our construction: GeFL-F's fraction of the training set [8] is divided over the classes by the profile of Section 3 with $\rho = 100$ (GeFL's own splits are balanced). The fraction is GeFL-F's on CIFAR-10 (0.5). GeFL does not use CIFAR-100; there we take 0.22, which gives the usual CIFAR-100-LT shape. No class receives more images than remain after the validation images are held out. On CIFAR-10-LT and CIFAR-100-LT the profile asks for more than that for the largest classes, which are therefore capped. So the realised ratio between the largest and the smallest class is below 100 (Table 3); we name every setting by its ρ. The class index orders the classes

Table 3: Datasets as used: classes, training images held by the clients, test images, and the realised ratio between the largest and the smallest class in the federation (natural imbalance: the dataset's own class sizes).

Dataset	Classes	Training images	Test images	Largest / smallest class
CIFAR-10-LT	10	18,485	10,000	48
CIFAR-100-LT	100	10,926	10,000	96
DermaMNIST	7	6,231	2,005	75
OrganAMNIST	11	19,994	17,778	5
BloodMNIST	8	10,359	3,421	3
PathMNIST	9	19,997	7,180	2
EMNIST (ByClass)	62	19,970	116,323	20
FER-2013	7	19,997	3,589	17

from the most to the least frequent, so the tail classes are the same in every run.

Natural imbalance. DermaMNIST [53], OrganAMNIST [53], BloodMNIST [53], PathMNIST [53], EMNIST (ByClass) [54] and FER-2013 [55] keep their own class sizes, with training sets capped at 20,000 images. Their classes are relabelled by training frequency. On three of them the realised ratio is small (Table 3): their results speak to label skew more than to a long tail.

Splits. The training images are split among $K = 10$ clients by a Dirichlet draw with $\alpha = 0.5$ per class. The draw is repeated, at most 100 times, until every client holds at least ten images; if it still fails, random images move from the largest to the smallest client. Client k trains head architecture k, so in the main setting each architecture has one client.

A held-out block of training images that no client holds forms a validation set: $\min(200, \lfloor 2000/C \rfloor)$ images per class, and at most a quarter of a class with natural imbalance. It has two uses: it trains the classifier that scores generated features in the diagnostics (Appendix C.1), and it sets the public bounds by which secure aggregation scales the released vectors. It is never used to select a model or a hyperparameter. The other unused training images are discarded. Every result is on the official test set. We checked that the class counts of every client are identical across methods for every dataset and seed.

6.4. Protocols

Main setting: different client models. The main setting is GeFL-F's setting with $K = 10$ clients, one per architecture, at $\rho = 100$ and Dirichlet $\alpha = 0.5$. The ratio sets the target profile; where a frequent class has fewer images than the profile asks for, it keeps all of them, so the realised ratio of the training set can be lower (Table 3, last column). The rounds of the three stages, T_{I}, T_{II} and T_{III}, are GeFL-F's: 50, 200 and 100 on CIFAR-10, and the same on CIFAR-100 (Appendix B lists every dataset). All federated methods of this setting share the same Stage-I extractor for a given dataset and seed: Stage I is computed once and reused.

One shared model. We follow the FedLTLib protocol [28], whose training settings come from CReFF [9]: CIFAR-10-LT and CIFAR-100-LT with the standard profile $n_c = \lfloor n_{\max} \rho^{-c/(C-1)} \rfloor$ at $\rho = 100$, where $n_{\max}$ is the number of images per class left after 200 per class (20 on CIFAR-100) are held out for validation. Rounding down gives the smallest CIFAR-100 class 4 images at $\rho = 100$.

There are 20 clients, Dirichlet $\alpha = 0.5$, and 8 clients (40%) take part in each round. All share one ResNet-8 [56], trained for 200 rounds of 10 local epochs with batch size 32 and SGD (learning rate 0.1, momentum 0.9, decay 0.99 per round). The baselines run the authors' LongTailLib training code unchanged, on our splits, with the backbone set to ResNet-8 for every method (the library defaults to ResNet-18 or ResNet-20 for FedLoGe, Fed-GraB, FedYoYo and FedETF); we evaluate them every fifth round and score Fed-GraB with its trained classifier.

For FedECS-LT the ResNet-8 is split after its second stage into a frozen extractor and a head. The extractor is trained for 100 rounds with the protocol's local updates, the generator for 100 rounds, and the head for 100 rounds with 10 real epochs per round; the learning-rate decay continues from the extractor's rounds. So FedECS-LT uses the baselines' 200 training rounds plus 100 generator rounds. Its head trains on frozen features of images without augmentation, whereas the baselines train end to end with augmentation. With one architecture, the consensus soft label is the current shared model's own prediction. FedECS-LT averages the clients' models with equal weights, as in the main setting; the benchmark's FedAvg weights them by their numbers of samples. The clients of each round are a seeded subset; the library draws its own, so participation is not paired between FedECS-LT and the baselines.

6.5. Methods compared

Table 4 lists every method, its reference and the origin of its code, and Table 5 what each method of the main setting exchanges and computes. The methods fall into these groups.

- **GeFL-F**, the method FedECS-LT extends: the difference to it answers RQ2 directly.
- **FedDF** [30], ensemble distillation on the server. GeFL distils on public data; we run FedDF's distillation on GeFL-F's generated features instead, so that no method uses extra data.
- **Per-client long-tail losses**: Balanced Softmax [18], FedLC [17] and FedRS [19]. They test whether a correction of the loss alone repairs the long tail. They are applied to GeFL-F without its generator.
- **Model-heterogeneous methods from HtFLlib** [14]: prototypes (FedProto, FedTGP), a generator of embeddings (FedGen), a shared global head (FedGH), shared last layers (LG-FedAvg) and shared subnetworks (FedMRL). They represent the other ways in which clients with different models exchange knowledge.
- **References**: Local, in which each client trains its extractor and head alone, is the reference without federation.

Methods designed for one shared model (Table 4, lower part) are compared in the one-shared-model setting only.

Settings and tuning. No baseline was tuned on our data; their settings come from different sources:

- the HtFLlib methods use the values of HtFLlib's run scripts, with 5 local epochs per round to match Stage III (Appendix B);
- the LongTailLib methods use the benchmark's common settings, which differ from some methods' own papers;
- Balanced Softmax and FedLC ($\tau = 1$) are re-implemented losses with the client's add-one smoothed class counts, and FedRS ($\alpha = 0.5$) with the classes the client lacks;

Table 4: Methods compared, their protocol and the origin of their code. HtFLlib [14] (Apache-2.0) methods are ported to Flower with HtFLlib's settings and 5 local epochs per round, on the shared frozen extractor. LongTailLib [28] methods run in the authors' code, downloaded at a fixed commit.

Method	Protocol	Code
GeFL-F [8]	different models	re-implementation of the authors' algorithm
FedDF [30]	different models	re-implementation: server distillation on generated features
Balanced Softmax [18]	different models	loss on each head, no generator
FedLC [17]	both	loss on each head, no generator / LongTailLib
FedRS [19]	different models	loss on each head, no generator
FedProto [3]	different models	HtFLlib
FedTGP [4]	different models	HtFLlib
FedGen [5]	different models	HtFLlib
FedGH [6]	different models	HtFLlib
LG-FedAvg [31]	different models	HtFLlib
FedMRL [7]	different models	HtFLlib
Local	different models	each client alone
FedAvg [1], FedProx [45], MOON [46]	one shared model	LongTailLib
CCVR [22], CReFF [9], FedETF [34]	one shared model	LongTailLib
Fed-GraB [10], FedLoGe [11]	one shared model	LongTailLib
FedYoYo [12], RUCR [13]	one shared model	LongTailLib
FedECS-LT	both	ours (Flower)

Table 5: Training budget and exchange of each method in the main setting: generator and its rounds (CIFAR), synthetic / real epochs per Stage-III round, what is exchanged in Stage III beyond the averaging of heads of the same architecture (which exchanges nothing at $K = 10$), whether the exact readout is applied (as the last step of FedECS-LT, or added as ER), and the loss of the real phase. Stage I is shared by all federated methods.

Method	Generator (rounds)	Synthetic / real epochs	Exchange in Stage III	Exact readout	Loss (real phase)
FedECS-LT	anchored or CCWA (200)	10 / 5	consensus soft labels	yes	logit-adjusted
GeFL-F	class rows, flat (200)	1 / 5	none at $K = 10$	yes	cross-entropy
FedDF	GeFL-F's (200)	1 / 5	server distillation	yes	cross-entropy
Balanced Softmax, FedLC, FedRS	none	0 / 5	none at $K = 10$	yes	per-client correction
HtFLlib methods	FedGen: in every batch	0 / 5	prototypes, heads or sub-networks	no	method's own
Local	none	alone: $5T_{\mathrm{I}} + 5T_{\mathrm{III}}$ epochs	none	no	cross-entropy

- FedDF uses 20 server distillation steps per round (Appendix B).

The settings of FedECS-LT (Table 2) were fixed during its development, on earlier runs of the same benchmarks with seeds 0 to 5. These included seeds 3 to 5 with the same client splits as the reported runs, and every earlier run was evaluated on the same test sets. The development compared about 36 configurations of the method and its variants on CIFAR-10-LT and 17 on CIFAR-100-LT (listed in the experiment ledger of the released code), and runs on MNIST, Fashion-MNIST and SVHN; it never used the datasets with natural class imbalance. The settings were not changed for the runs reported here (Section 6.9).

6.6. Metrics

Let r_{gc} be the recall of class c by the model of architecture g on the official test set. The test set is balanced on CIFAR-10 and CIFAR-100 and imbalanced on the naturally imbalanced datasets; balanced accuracy weights every class equally in either case. The primary metric is the balanced accuracy, $\frac{1}{G}\sum_g \frac{1}{C}\sum_c r_{gc}$: the mean over the $G = 10$ client models of each model's mean class recall. GeFL [8] also averages over the ten models, with top-1 accuracy at each model's best round. It is reported for every method without the readout (FedECS-LT with its readout, and its heads only) and, for the methods of the GeFL-F pipeline, also with the readout added; the second set answers RQ3.

The other metrics are:

- the recall of the tail classes, the last 40% by frequency rank, and of the worst class, $\min_c \frac{1}{G}\sum_g r_{gc}$, the lowest class recall after averaging recall over the models;
- the number of classes with zero recall for every model;
- the expected calibration error with 15 bins [57];
- the balanced accuracy of the weakest architecture;
- the accuracy on each client's own class mix: the test accuracy reweighted to the class frequencies of that model's clients.

Every per-class metric is logged for every class at every evaluation; Appendix C defines the secondary metrics of its per-dataset tables.

GeFL's code reports each model's best test accuracy over the evaluated rounds. We report the model after the last round, which involves no selection on the test set. GeFL's measure is given in Table C.16.

Cost is the communication per client, the wall-clock time of a run and the floating-point operations (Appendix B). The times were measured on a T4 GPU that in most notebooks ran a second run at the same time, so they are only indicative.

6.7. Protocol and implementation

Every configuration is run with seeds 3, 4 and 5; settings with fewer completed seeds say so. A seed fixes the split and the initialisation of every method, so runs with the same seed are paired.

FedECS-LT and every method of the main setting run in Flower [27]. The one-shared-model baselines run in the authors' LongTailLib code on the same client splits and are evaluated on the same test set. Appendix B gives the software and hardware.

All releases of FedECS-LT use Flower's SecAgg+ [21], with 30-bit stochastic quantisation of every vector after division by a public bound: a multiple of the federation size and of the largest value on the validation features for the class and readout sums, and fixed bounds for the counts, the CCWA rows and the consensus predictions. Values beyond eight times the bound would be clipped.

The Flower implementation was checked against our earlier implementation, which follows GeFL's code (Appendix B).

GPU computation is not bit-reproducible (cuDNN autotuning and non-deterministic GPU kernels), so two runs with the same seed differ. A notebook run twice with the same run list (BloodMNIST, 39 runs) measures this. The standard deviation of the difference between the two runs of one configuration is 1.2 points of balanced accuracy as trained and 0.6 with the exact readout.

6.8. Statistical analysis

We report the mean ± sample standard deviation over seeds. A difference to FedECS-LT is the mean paired difference over the seeds both methods have, with a 95% t interval and the paired t-test. The p-values are corrected by Holm's procedure [58] over the other methods of the same table and setting: every method without the readout, and the methods of the GeFL-F pipeline with the exact readout. Local is included; the pooled-data reference and the diagnostic runs and the heads of FedECS-LT alone are not. There is no correction across datasets or metrics. "Best" and "strongest other method" are decided on the seeds both methods have.

With three seeds the tests have little power, and they assume normally distributed paired differences: a sign-flip permutation test cannot give $p < 0.25$ with three paired differences. A test result near $p = 0.05$ can also depend on which methods form the Holm family. GPU non-determinism alone gives the mean of three paired differences a standard deviation of about half the two-run value above; we treat differences smaller than twice that, 1.4 points as trained and 0.7 with the exact readout, as unresolved. Exceeding the floor is necessary, not sufficient: a difference above it is resolved only if its paired test is also significant after Holm's correction.

6.9. Threats to validity

Internal validity.

- **Tuning.** The settings of FedECS-LT were fixed during its development on the same benchmarks, partly with the same seeds and client splits and on the same official test sets (Section 6.5), while the baselines were not tuned. The reported numbers may therefore favour FedECS-LT. There are no client splits that the development never saw.
- **Compute.** FedECS-LT trains its heads for ten synthetic epochs per round, against one for GeFL-F and FedDF and none for the methods without a generator (Table 5), and it communicates more (Table 12); the methods were not compared at matched communication or wall-clock time. How much of the result of FedECS-LT depends on the longer phase is not measured: the ablation with one synthetic epoch is not part of the results reported here. What GeFL-F or FedDF would reach with ten synthetic epochs is not measured either.
- **Missing runs.** Some settings have fewer than three completed seeds; their tables say so, and a value from one seed has no interval. With one shared model, runs of FedECS-LT failed on some seeds and imbalance ratios where the baselines finished, so the missing runs are not random; Table 11 gives the seeds that completed.
- **Ports.** The HtFLlib methods were designed for end-to-end training and run here on a frozen extractor with HtFLlib's settings. Section 7.1 shows that FedProto and FedTGP lose accuracy after their first evaluated rounds, which may mean that these settings do not suit frozen features.
- **Non-determinism.** Pairing by seed removes the variation of the split but not that of GPU computation (Section 6.7).

Construct validity.

- Balanced accuracy is a mean over ten models of different sizes, so a large gain of the small heads and a small gain of the large heads give the same number. Table C.22 gives the weakest architecture.

- The readout was added to the methods of the GeFL-F pipeline but not to the HtFLlib methods, so the margins of FedECS-LT over the latter include the readout's gain. The readout is the same linear classifier for every body, so accuracy with it does not measure every property of a representation.
- The server holds a small validation set (held out from the training data) for the generator diagnostics and the secure-aggregation bounds; a deployment without such data would need another source for the bounds.
- The last-round model can be below a method's best round; Table C.16 gives the best round.

External validity.

- The long tail is synthetic on the benchmarks, with the same tail classes in every run.
- The limits of the study's scope (a small frozen extractor, label shift only, one client per architecture) are listed in Section 8.

7. Results

7.1. Main comparison: different client models

Table 6 gives the balanced accuracy of every method on the long-tailed benchmarks (CIFAR-10 and CIFAR-100): FedECS-LT with its readout, its heads before the readout (heads only), and every other method without the readout; Table 7 gives the paired difference of FedECS-LT to each method. FedECS-LT has the highest balanced accuracy on both benchmarks. Its margin over the strongest other method is +15.1 points on CIFAR-10-LT (over FedDF) and +8.6 on CIFAR-100-LT (over FedDF), with Holm-corrected $p = 0.0052$ and 0.0042. Its heads alone are +5.2 and +4.3 points ahead of the strongest other method (FedDF and FedDF; $p = 0.066$ and 0.021). The readout accounts for the rest; whether the rest of the method still helps once every method has the readout is examined below. On CIFAR-100-LT, 0 classes on average are recognised by none of the ten client models of FedECS-LT (0.3 before the readout), against 12.3 for FedDF. Against GeFL-F, the method it extends, the difference is +18.9 points [+16.3, +21.6] on CIFAR-10-LT and +11.0 [+10.2, +11.8] on CIFAR-100-LT.

Other model-heterogeneous methods. The model-heterogeneous methods that exchange prototypes or shared sub-networks (FedProto, FedTGP, FedGH, LG-FedAvg, FedMRL) were designed for label skew without a global long tail; under the long tail their balanced accuracy is close to or below that of Local; none of them gives an architecture information about the classes its clients lack. FedProto, FedTGP and FedGen are best in the first rounds and lose 5.7 ± 1.0, 6.7 ± 0.7 and 5.3 ± 0.9 points of accuracy by the last round on CIFAR-10-LT (2.8 ± 0.5, 3.2 ± 0.3 and 3.1 ± 1.2 on CIFAR-100-LT); they are also the weakest on their clients' own class mixes, on which HtFLlib scores such personal models (Table C.20). The heads of FedECS-LT, too, are past their best round at the last one, by 3.4 ± 1.2 points on CIFAR-10-LT and 0.9 ± 0.3 on CIFAR-100-LT; Table C.16 gives every method's best round, as GeFL reports it (the readout is computed once, after the last round).

Accuracy on the clients' own class mixes. On the clients' own class mixes, FedECS-LT is below most other federated methods, and lower with its readout than with its heads only. There FedECS-LT reaches 59.2 ± 1.8 on CIFAR-10-LT and 28.3 ± 0.3 on CIFAR-100-LT (heads only: 68.6 ± 1.9 and 31.5 ± 0.3), below 11 of the 11 other federated methods on CIFAR-10-LT and 7 of 11 on CIFAR-100-LT; GeFL-F, Balanced Softmax, FedLC, FedRS, FedDF, Local, FedGH and FedMRL are above it on both (Table C.20).

Figure C.5 (Appendix C) shows the paired differences of the heads as trained on CIFAR-10-LT, before the readout of FedECS-LT, with their confidence intervals.

The same readout for every method. The readout uses only a head's body, so it can be added to every method of the GeFL-F pipeline. Table 8 compares these methods with every last layer set by the same readout, so that the remaining differences are those of the bodies, which the other components of FedECS-LT train. With the same readout the gap to every method shrinks: FedECS-LT remains +2.3 points ahead of GeFL-F on CIFAR-10-LT and +3.8 on CIFAR-100-LT, and +1.4 and +1.5 points ahead of the strongest other method (Balanced Softmax and FedRS). The differences to the strongest other method are not significant after Holm's correction ($p = 0.37$ and 0.058), although on CIFAR-100-LT the uncorrected 95% interval [+0.6, +2.4] excludes zero (CIFAR-10-LT: [-0.7, +3.6]). On CIFAR-100-LT the differences to GeFL-F and FedDF remain significant with the readout ($p = 0.0095$ and 0.014). Most of the margin over the published methods is therefore the

Table 6: Balanced accuracy (%) on the test set; mean ± standard deviation over seeds 3–5. FedECS-LT with its exact readout; FedECS-LT (heads only): its heads before the readout; every other method without the readout (heads as trained). Long-tailed benchmarks (imbalance ratio 100), 10 clients with ten different head architectures, Dirichlet label split ($\alpha = 0.5$). Best in each column in bold.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**52.6 ± 1.3**	**18.7 ± 0.2**
FedECS-LT (heads only)	42.8 ± 2.0	14.3 ± 0.6
GeFL-F	33.7 ± 1.2	7.7 ± 0.4
Balanced Softmax	36.3 ± 1.2	9.0 ± 0.3
FedLC	34.3 ± 1.1	8.5 ± 0.3
FedRS	34.1 ± 1.1	8.5 ± 0.3
FedDF	37.5 ± 1.2	10.0 ± 0.5
Local	32.1 ± 0.7	7.9 ± 0.8
FedProto	27.1 ± 0.8	4.8 ± 0.3
FedTGP	25.3 ± 1.2	4.5 ± 0.6
FedGen	26.9 ± 2.0	4.5 ± 1.3
FedGH	31.9 ± 1.1	6.5 ± 0.1
LG-FedAvg	31.3 ± 0.8	6.3 ± 0.1
FedMRL	33.7 ± 1.0	8.3 ± 0.2

readout's, and three seeds do not show whether the other components of FedECS-LT also train a better body than the strongest other method.

7.2. Rare classes

Table 9 reports the recall of the tail classes, and Fig. C.6 (Appendix C) the recall of every class against its number of training images. Against GeFL-F, the gain of FedECS-LT on CIFAR-10-LT is in the tail: its recall rises by +36.4 points on the tail classes and by +2.7 on the head classes. On CIFAR-100-LT the gain is spread over all three groups (head +12.6, medium +15.4, tail +6.5 points); its tail recall is 8.2 ± 0.2 against 1.7 ± 0.1, and its worst-class recall 0.2 ± 0.2 against 0.0 ± 0.0. Against FedDF, FedECS-LT gains on the tail classes (+39.7 points on CIFAR-10-LT) and loses on the head classes (-8.4). With the exact readout the tail gain over Balanced Softmax is -0.03 on CIFAR-10-LT and -0.1 on CIFAR-100-LT: the tail gain over the loss-only methods comes from the last layer (Table C.19). Figure C.7 (Appendix C) splits the recall into the head, medium and tail classes.

Table 7: Balanced accuracy of FedECS-LT (with its readout) minus each method (points), paired over seeds 3–5, with the 95% t confidence interval; bold where the paired t-test is significant after Holm's correction over the methods ($p < 0.05$). Same setting as Table 6.

Method	CIFAR-10	CIFAR-100
GeFL-F	**+18.9 [+16.3, +21.6]**	**+11.0 [+10.2, +11.8]**
Balanced Softmax	**+16.3 [+13.5, +19.0]**	**+9.7 [+8.9, +10.5]**
FedLC	**+18.3 [+16.2, +20.5]**	**+10.2 [+9.4, +11.0]**
FedRS	**+18.5 [+16.4, +20.6]**	**+10.2 [+9.5, +11.0]**
FedDF	**+15.1 [+12.6, +17.5]**	**+8.6 [+7.3, +10.0]**
Local	**+20.5 [+17.2, +23.9]**	**+10.7 [+8.6, +12.8]**
FedProto	**+25.5 [+24.2, +26.8]**	**+13.9 [+13.5, +14.4]**
FedTGP	**+27.3 [+25.4, +29.2]**	**+14.2 [+13.0, +15.4]**
FedGen	**+25.7 [+23.4, +28.1]**	**+14.1 [+10.5, +17.8]**
FedGH	**+20.7 [+18.1, +23.3]**	**+12.1 [+11.8, +12.5]**
LG-FedAvg	**+21.3 [+19.2, +23.4]**	**+12.4 [+12.1, +12.7]**
FedMRL	**+18.9 [+16.8, +21.0]**	**+10.4 [+9.9, +10.9]**

7.3. Datasets with natural class imbalance

Table 10 repeats the comparison on six datasets whose class sizes are their own (DermaMNIST, OrganAMNIST, BloodMNIST, PathMNIST, EMNIST (ByClass) and FER-2013). FedECS-LT has the highest mean balanced accuracy on all six, with margins of +6.4 (OrganAMNIST) to +18.4 points (PathMNIST) over the strongest other method without the readout; the margin is significant after Holm's correction on three of the five datasets with three seeds (EMNIST (ByClass) has one seed). These datasets were not used in the development of FedECS-LT. With the same readout for every method of the GeFL-F pipeline the margins are -0.03 (BloodMNIST, against FedDF) to +0.9 points (FER-2013). None of them is significant after Holm's correction on the datasets with three seeds (the smallest p is 0.051, on OrganAMNIST); all but the smallest exceed the noise floor of 0.7 points of Section 6.8, which is necessary but not sufficient for a resolved difference. On one of the datasets a method with the readout has the higher mean (Table C.17). The readout changes the balanced accuracy of FedECS-LT by -0.3 points on OrganAMNIST and by +11.6 on PathMNIST. The separation gate chose the generator with class rows on FER-2013, where the class means are too close for an anchor.

Table 8: Balanced accuracy (%) with every architecture's last layer set by the exact readout (Section 5.4), the last step of FedECS-LT, added to every method; mean ± standard deviation over seeds 3–5. Methods of the GeFL-F pipeline (same extractor, heads with a CNN body and a linear last layer); the HtFLlib methods and Local were not evaluated with the readout. Same setting as Table 6.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**52.6 ± 1.3**	**18.7 ± 0.2**
GeFL-F	50.3 ± 1.2	14.9 ± 0.2
Balanced Softmax	51.2 ± 0.7	17.2 ± 0.3
FedLC	51.2 ± 0.7	17.2 ± 0.3
FedRS	51.1 ± 0.7	17.2 ± 0.3
FedDF	51.0 ± 1.2	15.8 ± 0.3

7.4. One shared model

Table 11 gives the results in the FedLTLib protocol. Here several methods are stronger: FedYoYo, FedLoGe, CCVR, FedETF, FedAvg and MOON are above FedECS-LT on CIFAR-10-LT and FedYoYo, FedLoGe and CCVR on CIFAR-100-LT, and 5 and 8 methods are below it. At imbalance ratio 100, FedECS-LT reaches 58.4 ± 1.6 on CIFAR-10-LT and 34.0 on CIFAR-100-LT, against 61.0 ± 1.8 and 33.2 ± 0.3 for FedAvg, while the strongest method, FedYoYo, is 19.8 and 9.0 points ahead of FedECS-LT on the seeds both have (seeds 3 and 4 on CIFAR-10-LT and 4 on CIFAR-100-LT; Table 11 averages every seed of each method). A possible explanation is that with one shared model the baselines train the whole network end to end on real, augmented data, whereas FedECS-LT trains its head on frozen features (Section 6.4). We did not measure the representations in this protocol, so this explanation is not tested. Here the readout does not help: relative to the heads of FedECS-LT it changes the balanced accuracy by -3.0 points on CIFAR-10-LT and -0.1 on CIFAR-100-LT. With one shared head trained by every client, the last layer is not fitted to one client's label prior, the failure the readout addresses; we did not test further why it is lower.

Comparison with published numbers. Our baselines run in the authors' code on our splits and report the model after the last round. Against FedYoYo's evaluation in the same protocol [12], most methods are within about three points of the published values (e.g. FedAvg 58.2, CCVR 68.1, FedETF 68.3, FedYoYo 81.5 on CIFAR-10-LT). FedLTLib [28] reports the best round and

Table 9: Recall (%) of the tail classes (the 40% least frequent); FedECS-LT with its readout, other methods without the readout (as in Table 6); mean ± standard deviation over seeds 3–5. Same setting as Table 6.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**50.3 ± 1.5**	**8.2 ± 0.2**
FedECS-LT (heads only)	31.4 ± 2.7	6.4 ± 0.3
GeFL-F	13.9 ± 1.2	1.7 ± 0.1
Balanced Softmax	20.0 ± 1.9	2.2 ± 0.1
FedLC	16.1 ± 1.4	1.7 ± 0.1
FedRS	15.7 ± 1.4	1.7 ± 0.1
FedDF	10.6 ± 1.0	0.8 ± 0.1
Local	14.0 ± 1.2	1.6 ± 0.2
FedProto	10.7 ± 1.9	0.9 ± 0.0
FedTGP	10.8 ± 1.7	1.2 ± 0.5
FedGen	8.6 ± 1.1	0.7 ± 0.3
FedGH	13.3 ± 0.8	1.1 ± 0.1
LG-FedAvg	13.3 ± 0.6	1.0 ± 0.0
FedMRL	14.9 ± 1.2	1.7 ± 0.1

gives higher values than ours for every method. Three methods are further below the published values on CIFAR-10-LT in our runs: CReFF (70.5 in [12]), Fed-GraB (70.6) and FedLC (69.0). They run in the benchmark's code with its common settings (learning rate 0.1, batch size 32, ten local epochs), which differ from those in the methods' own papers; we did not tune them and report them as obtained. FedYoYo's evaluation trains 300 rounds and ours 200. Figure C.8 (Appendix C) shows the balanced accuracy over the rounds.

7.5. Representations and generated features

Figure 4 shows the test embeddings of GeFL-F and FedECS-LT on the same images. The embeddings of FedECS-LT separate the classes slightly better: the leave-one-out 10-nearest-neighbour accuracy of the embeddings, which does not depend on the map, is 40.9% for FedECS-LT, 37.7% for GeFL-F and 38.9% for FedDF (mean over the ten architectures and seeds 3–5). It is higher for FedECS-LT in 29 of the 30 architecture–seed pairs against GeFL-F and in 24 against FedDF. Figure C.9 (Appendix C) overlays real and generated features.

Table 10: Balanced accuracy (%) on datasets with natural class imbalance (FedECS-LT with its readout, other methods without the readout, as in Table 6) (training sets capped at 20,000 images); mean ± standard deviation over seeds 3–5 (a value without ± is from one seed). 10 clients with ten different head architectures, Dirichlet $\alpha = 0.5$.

Method	DermaMNIST	OrganAMNIST	BloodMNIST	PathMNIST	EMNIST (ByClass)	FER-2013
FedECS-LT	**45.4 ± 1.9**	**83.3 ± 1.3**	**86.3 ± 1.3**	**73.6 ± 0.4**	**67.3**	**39.2 ± 1.0**
FedECS-LT (heads only)	36.8 ± 0.8	83.7 ± 2.5	84.2 ± 2.0	61.9 ± 3.2	62.1	30.9 ± 1.5
GeFL-F	27.3 ± 0.3	76.9 ± 3.5	78.5 ± 2.2	55.2 ± 2.3	49.8	27.5 ± 0.2
Balanced Softmax	28.6 ± 0.1	68.1 ± 4.8	69.9 ± 3.8	54.6 ± 2.2	45.6	26.6 ± 0.4
FedLC	26.0 ± 0.7	63.9 ± 4.4	66.5 ± 3.9	51.9 ± 2.6	42.1	25.7 ± 0.2
FedRS	26.0 ± 0.6	63.5 ± 4.4	66.0 ± 3.9	52.6 ± 1.4	41.4	25.7 ± 0.3
FedDF	20.5 ± 1.4	64.7 ± 6.1	70.6 ± 7.5	48.2 ± 1.2	47.2	25.7 ± 0.3
Local	25.3 ± 0.5	63.5 ± 4.2	63.6 ± 3.6	49.3 ± 3.4	41.4	25.2 ± 0.5
FedProto	25.5 ± 1.5	58.6 ± 4.1	65.4 ± 4.5	54.0 ± 0.8	39.4	20.8 ± 0.5
FedTGP	24.7 ± 1.0	50.7 ± 3.9	59.5 ± 2.4	49.8 ± 1.3	34.8	20.6 ± 0.7
FedGen	26.0 ± 0.6	60.1 ± 5.1	64.1 ± 3.2	51.7 ± 0.5	32.2	24.4 ± 0.5
FedGH	27.1 ± 0.2	61.8 ± 4.2	65.9 ± 3.6	53.8 ± 1.2	38.7	24.7 ± 0.4
LG-FedAvg	26.8 ± 0.4	62.6 ± 4.3	65.9 ± 3.2	53.3 ± 1.2	38.5	24.7 ± 0.6
FedMRL	27.1 ± 0.3	64.2 ± 4.3	67.2 ± 3.5	53.7 ± 1.0	42.1	25.9 ± 0.5

7.6. Calibration, fairness across clients and cost

The expected calibration error of FedECS-LT is 21.2 ± 0.5 on CIFAR-10-LT against 48.0 ± 2.2 for GeFL-F and 35.9 ± 1.9 for FedDF, the strongest other method in balanced accuracy; it ranks 3 of thirteen methods on CIFAR-10-LT and 5 on CIFAR-100-LT, where its error is 43.8 ± 0.4 against 42.4 ± 0.7 for GeFL-F (Table C.21); the weakest architecture of FedECS-LT reaches 48.5 ± 1.5 balanced accuracy (Appendix C). Table 12 reports the cost on CIFAR-10-LT. FedECS-LT takes 1.13 times the wall-clock time of GeFL-F and 1.30 times its communication. Against Balanced Softmax, which trains its heads locally on real features, FedECS-LT communicates 185 times as much and takes 15 times as long; with the readout for both, it is +1.4 points ahead.

8. Limitations

The threats to the validity of the study design are discussed in Section 6.9; this section lists the limits of what the study covers.

- The extractor is small and frozen, as in GeFL-F, and shared by all clients, so the clients' architectures differ only in their heads; we did not study pre-trained or larger extractors, or clients with different backbones.

Table 11: Balanced accuracy (%) with one ResNet-8 shared by all clients (FedLTLib protocol: 20 clients, 40% per round, 200 rounds, Dirichlet $\alpha = 0.5$); mean ± standard deviation over the seeds that completed (seeds 3–5; a value without ± is from one seed). Baselines in the authors' LongTailLib code on the same client splits, evaluated after the last round; FedECS-LT with its readout and, separately, its heads only. Fed-GraB: accuracy of its own trained classifier as logged by the library.

Method	CIFAR-10	CIFAR-100
FedECS-LT	58.4 ± 1.6	34.0
FedECS-LT (heads only)	61.3 ± 1.8	34.1
FedLC	58.3 ± 1.9	32.8 ± 0.1
FedAvg	61.0 ± 1.8	33.2 ± 0.3
FedProx	54.4 ± 1.6	31.2 ± 0.4
MOON	59.9 ± 1.9	33.4 ± 0.2
CCVR	68.0 ± 1.2	35.8 ± 0.1
CReFF	56.8 ± 2.3	28.1 ± 0.0
FedETF	65.3 ± 4.3	32.8 ± 0.2
Fed-GraB	49.4 ± 2.4	27.8 ± 0.1
FedLoGe	66.7 ± 2.9	39.4
FedYoYo	**78.1 ± 0.9**	**42.6 ± 0.5**
RUCR	54.0 ± 1.6	32.9 ± 0.2

- No published method addresses both different client models and a long tail, so the comparison is with long-tail losses applied per client, with FedDF and with model-heterogeneous methods designed for label skew; the long-tail methods CCVR and FedETF are not compared with different client models.
- The client splits are a pure label shift (Section 3); feature shift and concept shift between clients are not tested.
- In the main setting each architecture has one client ($K = 10$).
- The released class statistics are not differentially private (Section 5.5); FedECS-LT should not be called privacy-preserving beyond the guarantee of secure aggregation against an honest-but-curious server.
- FedECS-LT costs more than GeFL-F (Table 12): its extra time is mostly in the longer synthetic phase of Stage III, and its extra commu-

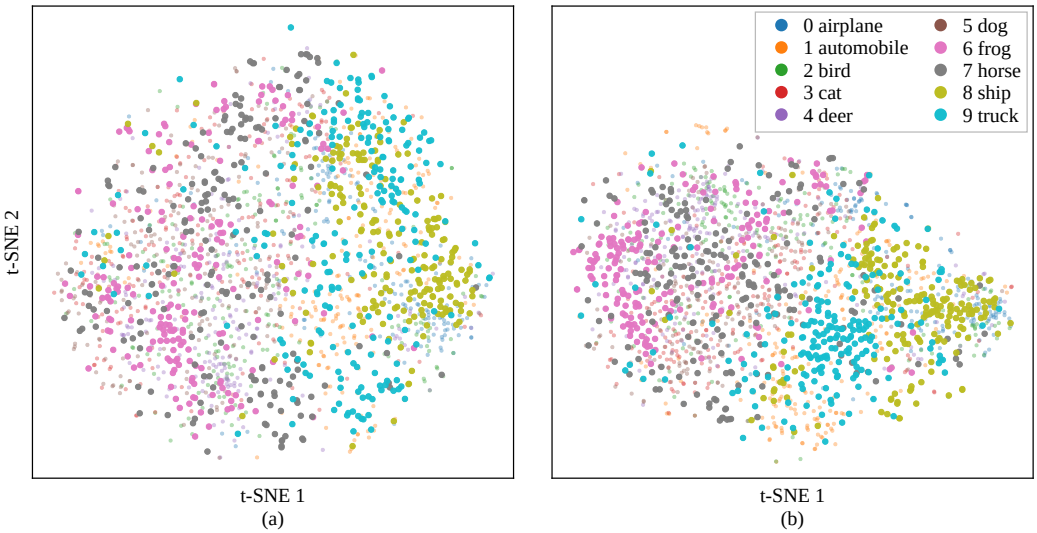


Figure 4: t-SNE [59] of the test embeddings of (a) GeFL-F and (b) FedECS-LT on the same 2,000 test images of CIFAR-10-LT (one architecture, seed 3; PCA to 50 dimensions, PCA initialisation, perplexity 30 [60]). Tail classes are drawn larger. Cluster sizes and distances between clusters are not interpretable [61]; only the grouping is.

nication mostly in the training of the generator, which with its projection of the class means is larger than GeFL-F's.

- The absolute accuracy of the main setting is low. GeFL's authors attribute GeFL-F's lower accuracy on CIFAR-10 to its feature extractor and target networks (their term for the heads) [8]. In GeFL's own setting (a balanced training set divided evenly among 10 clients) our earlier implementation of GeFL-F reaches 56.8% on CIFAR-10 (59.3% with GeFL's best-round measure; GeFL's Fig. 4 shows about 55.9%) and 12.5% on CIFAR-100 (Appendix B). These numbers are not comparable with centralised long-tail results for ResNet-32 and larger networks.

9. Conclusion

We examined three places where generator-based model-heterogeneous federated learning can lose rare classes: coupled weight decay under Adam on clients that lack a class, which shrinks the rare classes' rows in GeFL-F's generator but is not yet shown to cause their loss; a last layer fitted to one

Table 12: Cost on CIFAR-10-LT (imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$): communication per client over the whole training, wall-clock time of a run on a T4 GPU (in most notebooks shared with a second run), forward MFLOPs per image and training GFLOPs per round; mean over seeds 3–5.

Method	Comm./client (MB)	Time (s)	MFLOPs/img	Train. GFLOPs/round
FedECS-LT	11,278 ± 113	7,910 ± 2,253	**2.45**	1,424 ± 112
GeFL-F	8,659	7,025 ± 446	**2.45**	579 ± 42
Balanced Softmax	60.8	542 ± 99	**2.45**	**485 ± 34**
FedLC	60.8	501 ± 118	**2.45**	**485 ± 34**
FedRS	60.8	**417 ± 131**	**2.45**	**485 ± 34**
FedDF	8,659	5,050 ± 2,736	**2.45**	579 ± 42
Local	-	536 ± 98	-	-
FedProto	18.8	4,025 ± 109	2.45	485 ± 34
FedTGP	18.8	4,121 ± 218	2.45	485 ± 34
FedGen	128	4,443 ± 1,033	2.45	485 ± 34
FedGH	18.8	3,149 ± 103	2.45	485 ± 34
LG-FedAvg	18.8	2,890 ± 212	2.45	485 ± 34
FedMRL	**17.1**	4,545 ± 804	3.84	871 ± 34

client's label prior; and generated features that do not carry their label. Each is addressed by a component of FedECS-LT that uses class statistics computed by secure aggregation or, for the heads, logit adjustment and consensus soft labels. Which of these components carry the effect beyond the readout is not yet measured. FedECS-LT, which applies these repairs together and ends with an exact readout of every last layer, improves balanced accuracy over the published methods when clients have different models, on CIFAR-10-LT mostly through the recall of the tail classes. Most of the gain is the readout's: added to every method of the GeFL-F pipeline, it leaves FedECS-LT a lead over the strongest other method that three seeds do not resolve. With one model shared by all clients, 6 other methods are better on CIFAR-10-LT. FedECS-LT also trades accuracy on the clients' own class mixes and, for its readout and consensus, more communication and the condition that every client can evaluate every head.

CRediT authorship contribution statement

[Author 1]: Conceptualization, Methodology, Software, Investigation, Writing – original draft. [Author 2]: Supervision, Writing – review & editing. [Use the CRediT roles that apply to each author.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The code of FedECS-LT and of every compared method as run here, the notebooks, every run's logs and metrics and the data of every figure are available at [repository URL, e.g. a GitHub release with a DOI from Zenodo]. All datasets used are public (Section 6.3).

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used [name of the tool] to [e.g. write and test parts of the experiment code, draft and edit passages of the text]. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Proofs

Lemma 1. By the triangle inequality, $\|\frac{1}{K}\sum_k r_{k,c}\| \le \frac{1}{K}\sum_k \|r_{k,c}\| = \|r_c\| + \frac{1}{K}\Big(\sum_{k\in H_c}\Delta_k + \sum_{k\notin H_c}\Delta_k\Big)$, and the two sums are $m_c\bar{G}_c$ and $-(K-m_c)\bar{D}_c$. Equality holds if and only if all vectors are non-negative multiples of one another. □

Lemma 2. A non-holder's batches contain no sample of class c, so its data loss does not depend on r_c and the gradient of the regularised loss is λr_c. Adam's update $\eta\hat{m}/(\sqrt{\hat{v}}+\epsilon)$ is unchanged when the whole gradient sequence is multiplied by $a > 0$, up to the ϵ term, so the step does not depend on λ while $\lambda|r_{c,j}| \gg \epsilon$. A client keeps its Adam state across rounds while its parameters are reset to the global ones, so a single step can exceed η. □

The step ratio (Eq. (1)), a heuristic. Per coordinate an Adam step is $\eta\psi$ with $\psi = \hat{m}/\sqrt{\hat{v}}$. For a gradient of constant sign ψ tends to that sign; in general $|\psi| \le (1-\beta_1)/\sqrt{1-\beta_2}$, about 3.2 for the default β_1, β_2 [15]. A non-holder sees a decay gradient of constant sign, so $\psi \approx \operatorname{sign}(r_{c,j})$ on each of its q_k steps. Assume that a holder moves the row only on the $q_k f_{kc}$ steps whose batch contains class c, with $|\psi| \le 1$, and neglect its other steps. Under flat averaging the drift per round is $\Delta r_{c,j} = \frac{\eta}{K}\left[\psi_h \sum_{k \in H_c} q_k f_{kc} - \operatorname{sign}(r_{c,j}) \sum_{k \notin H_c} q_k\right]$, which points toward zero for every coordinate and every holder direction when $\kappa_c < 1$. The approximation neglects momentum carried between sparse holder steps and the drift of $\hat{v}$. □

Lemma 3. (1) $f + a_k = Wu(h) + (b + a_k)$, and $b \mapsto b + a_k$ is a translation with identity Jacobian, so the gradients agree at corresponding points; induct over the steps. (2) $\alpha f_c = \tilde{w}_c^\top u + \tilde{b}_c$ with $\tilde{w}_c = \alpha w_c$, $\tilde{b}_c = \alpha b_c$; the gradient in $\tilde{w}_c$ is $1/\alpha$ times that in w_c, so a step of size η on w_c is a step of $\eta\alpha^2$ on $\tilde{w}_c$, and the body sees the same logits. □

Lemma 4. Addition is associative and commutative. In particular the class-balanced linear discriminant of Eq. (8) is computed exactly by one round of secure aggregation. □

Lemma 5. $\frac{1}{C}\sum_c(\mathrm{LSE}(f) - f_c) = \mathrm{LSE}(f) - \bar{f}$ and $\mathrm{KL}(u \| \mathrm{softmax} f) = -\log C + \mathrm{LSE}(f) - \bar{f}$; expanding LSE around constant logits gives the quadratic term. □

Lemma 6 (alternation, quadratic losses)**.** *Let the synthetic and real losses be quadratic with Hessians $H_s, H_r \succeq 0$ and minimisers θ_s, θ_r, and let $P_s = (I - \eta H_s)^{S_s}$, $P_r = (I - \eta H_r)^{S_r}$ for S_s synthetic and S_r real steps per round. Let $0 \le \eta\lambda \le 1$ for every eigenvalue λ of H_s and H_r. If $\|P_r P_s\| < 1$, the rounds converge to $\theta^\star = \theta_r + \Omega(\theta_s - \theta_r)$ with $\Omega = (I - P_r P_s)^{-1} P_r (I - P_s)$. For commuting Hessians the synthetic weight along eigendirection i is $\omega_i = \rho_{r,i}(1 - \rho_{s,i})/(1 - \rho_{r,i}\rho_{s,i}) \le \rho_{r,i}$, which equals one along directions that the real loss leaves free and the synthetic loss does not.*

Proof. One round is the affine map $\theta \mapsto P_r P_s \theta + P_r(I - P_s)\theta_s + (I - P_r)\theta_r$, a contraction when $\|P_r P_s\| < 1$; its fixed point has the stated form. With commuting Hessians all matrices are diagonal in a common eigenbasis. □

Lemma 7 (what a mean anchor can express)**.** *Let the class-c sampler be $X_c = \mathrm{ReLU}(\mu_c + g(Z, A\mu_c))$ with Z shared across classes and g L-Lipschitz in its second argument. Then $W_1(G_c, G_{c'}) \le (1 + L\|A\|_2)\|\mu_c - \mu_{c'}\|$.*

Proof. Couple both classes through the same Z; ReLU is 1-Lipschitz. □

In units of the root-mean-square within-class spread, two class means are on average $\sqrt{2FC/(C-1)}$ apart (over distinct pairs). When F is small the anchor carries little class information, and the factor $1 + L\|A\|_2$ is not controlled; this is why the gate of Section 5.2 falls back to class rows.

Lemma 8 (least-displacement moment correction)**.** *Among all maps T with $\mathbb{E}\,T(X) = \mu_c$ and $\mathbb{E}\|T(X) - \mu_c\|^2 = V_c$, where X has mean $\tilde{\mu}_c$ and spread $\tilde{V}_c$, the map $T^\star(x) = \mu_c + \sqrt{V_c/\tilde{V}_c}\,(x - \tilde{\mu}_c)$ minimises $\mathbb{E}\|T(X) - X\|^2$. With exact covariances instead of spreads the minimiser is the linear map attaining the Gelbrich bound [62].*

Proof. With $Y = T(X)$, $\mathbb{E}\|Y - X\|^2 = V_c + \tilde{V}_c + \|\mu_c - \tilde{\mu}_c\|^2 - 2\mathbb{E}\langle Y - \mu_c, X - \tilde{\mu}_c\rangle$, and the Cauchy–Schwarz inequality bounds the last expectation by $\sqrt{V_c\tilde{V}_c}$, with equality if and only if $Y - \mu_c = \sqrt{V_c/\tilde{V}_c}(X - \tilde{\mu}_c)$. The deployed map adds a final ReLU and clips the scale, so it matches the moments only approximately. □

Appendix B. Experimental details

Models. The extractor, the ten head architectures (CNN-1 to CNN-10) and GeFL-F's conditional VAE follow GeFL-F [8]; heads are trained by SGD with learning rate 0.1. The extractor is one convolutional block followed by 2×2 max pooling: one 3×3 convolution with three output channels on OrganAMNIST, EMNIST and FER-2013, and two convolutions with three and ten output channels on CIFAR-10, CIFAR-100, DermaMNIST, BloodMNIST and PathMNIST, each convolution followed by batch normalisation and ReLU. Rounds of the three stages ($T_{\mathrm{I}}/T_{\mathrm{II}}/T_{\mathrm{III}}$): 50/200/100 on CIFAR-10 and CIFAR-100; with natural imbalance, 20/100/50 on OrganAMNIST and EMNIST and 50/100/100 on DermaMNIST, BloodMNIST, PathMNIST and FER-2013.

Reproduction check. Before the comparison, the Flower implementation was checked against our earlier, single-process implementation (which follows GeFL's code) on MNIST-LT and CIFAR-10-LT with seed 3. The client splits were identical (every client's class counts), every client training step matched weight for weight in unit tests, and the balanced accuracy of GeFL-F and

FedECS-LT was within the larger of two points and three seed standard deviations of the earlier runs. The earlier CIFAR-10 reference of FedECS-LT was an older configuration, so its tolerance was three points.

Head architectures. CNN-1 to CNN-10 have convolutional blocks of widths (16), (16, 32), (20, 40), (10, 20), (16, 32, 64), (20, 40, 80), (10, 20, 40), (16, 32, 64, 128), (20, 40, 80, 100) and (10, 20, 40, 80); the widths do not increase monotonically from CNN-1 to CNN-10. Each block is a 3×3 convolution, batch normalisation (the naturally imbalanced datasets only), ReLU and 2×2 max pooling, followed by a linear last layer.

FedDF. After each Stage-III round the server distils every architecture for 20 SGD steps (batch 128, learning rate 0.05, no momentum) on fresh generated features with uniformly drawn labels; the target is the mean softmax of the ten architectures, and batch-normalisation statistics are kept. FedDF's original method distils on unlabelled public data instead.

GeFL's own setting. In GeFL's setting (balanced training data divided evenly among 10 clients, 3 seeds, our earlier implementation), GeFL-F reaches 56.8% on CIFAR-10 after the last round and 59.3% with GeFL's measure, against 55.9% read from GeFL's Fig. 4 [8], which reports each model's best round. An earlier configuration of FedECS-LT (one synthetic epoch per round and an older head training) reaches 57.8% and 60.0%. On CIFAR-100, which GeFL does not report, GeFL-F reaches 12.5% and that configuration 18.0%.

Evaluation. Models are evaluated every five rounds on the test set before and after aggregation; the final model is the one after the last round (no selection on the test or the validation set). Methods with personal heads (LG-FedAvg, FedGH, FedGen and the other HtFLlib methods) are scored with each client's head after its last local training; HtFLlib scores LG-FedAvg, FedGH and FedGen right after the server's update instead. FedProto and FedTGP classify by the distance to class prototypes; their probabilities for the calibration metrics are a softmax of the negative mean squared distances.

Cost. Communication is the bytes of the model and statistics messages per client, with a secure sum counted as eight bytes per summed value and client, without the protocol's key and share messages and without the test evaluation; the readout's sums are included. Time is the wall-clock time of

a run on a T4 GPU, which in most notebooks ran a second run at the same time. Computation is the forward floating-point operations per image, with training counted as three forward passes.

Software and hardware. FedECS-LT and every method of the main setting run in Flower 1.42 [27], a simulation with the Ray backend, one simulation per run. All runs used Kaggle notebooks with two NVIDIA T4 GPUs, in most notebooks with two runs per GPU at a time.

HtFLlib ports. Every client keeps the frozen Stage-I extractor and starts from its Stage-I head body, followed by a parameter-free pooling to 512 dimensions and a linear layer, as in HtFLlib's models. Training uses HtFLlib's run-script settings (SGD, learning rate 0.01, batch 10) with 5 local epochs per round for the rounds of Stage III; personal models persist across rounds. FedGen's label counts are the federation's class counts.

Appendix C. Further results

Per-dataset tables of every metric are generated from the same runs (Tables C.13 and C.14). Besides the metrics of Section 6.6 they give macro-F1, macro precision, the multi-class Matthews correlation coefficient [63], the geometric mean of the class recalls (each floored at 1%, computed for each model and then averaged) and macro AUC-PR; the class-wise form of the calibration error [64] is logged as well. The figures of this appendix show further views of the same runs (Section 7 cites each).

Appendix C.1. Fidelity of the generated features

After Stage I the server trains a referee classifier (CNN-5 head, Adam, 40 epochs) on the features of its validation images, which no client holds, and measures for each class the share of 300 generated features of that class that the referee assigns to it. Table C.15 gives the mean over the classes of each group. Chance is 10% on CIFAR-10-LT and 1% on CIFAR-100-LT. The referee's accuracy on held-out real features was not logged, so the values compare generators with each other, not with real data. The moment correction raises the head classes' fidelity of FedECS-LT on CIFAR-10-LT from 11.0 ± 2.2 to 23.5 ± 6.6 and lowers that of the tail classes from 44.1 ± 5.0 to 40.3 ± 2.8; on CIFAR-100-LT the mean is 8.8 ± 1.2 before and 8.8 ± 0.4 after. It matches the moments it is built for (Section 5.2), not the

Table C.13: CIFAR-10-LT (imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$): main metrics (%); FedECS-LT with its readout, its heads only, other methods without the readout; mean ± standard deviation over seeds 3–5. Tail: 40% least frequent classes; G-mean with recalls floored at 1%.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	**52.6 ± 1.3**	**52.3 ± 1.3**	**53.4 ± 1.3**	**47.4 ± 1.4**	**51.4 ± 1.3**	**50.3 ± 1.5**	**39.4 ± 1.9**	**56.0 ± 1.7**
FedECS-LT (heads only)	42.8 ± 2.0	40.9 ± 2.3	47.9 ± 1.2	37.3 ± 2.0	37.5 ± 2.7	31.4 ± 2.7	22.5 ± 2.2	45.5 ± 2.4
GeFL-F	33.7 ± 1.2	26.9 ± 1.4	38.3 ± 3.3	28.2 ± 1.4	15.7 ± 1.8	13.9 ± 1.2	4.8 ± 1.1	34.0 ± 2.2
Balanced Softmax	36.3 ± 1.2	30.4 ± 1.2	39.5 ± 2.7	30.9 ± 1.3	19.1 ± 1.6	20.0 ± 1.9	9.8 ± 1.2	36.3 ± 2.0
FedLC	34.3 ± 1.1	27.4 ± 0.9	40.2 ± 2.8	29.0 ± 1.2	15.2 ± 0.9	16.1 ± 1.4	6.2 ± 0.2	35.4 ± 1.8
FedRS	34.1 ± 1.1	27.2 ± 0.9	40.3 ± 2.9	28.9 ± 1.2	15.0 ± 0.9	15.7 ± 1.4	6.0 ± 0.3	35.3 ± 1.8
FedDF	37.5 ± 1.2	30.6 ± 1.4	43.7 ± 5.1	32.1 ± 1.3	18.3 ± 1.9	10.6 ± 1.0	1.6 ± 0.8	41.5 ± 1.9
Local	32.1 ± 0.7	25.0 ± 0.7	37.5 ± 2.3	26.8 ± 0.8	13.3 ± 0.6	14.0 ± 1.2	5.0 ± 0.7	33.9 ± 1.5
FedProto	27.1 ± 0.8	21.0 ± 0.4	27.6 ± 0.8	20.4 ± 0.8	12.3 ± 0.5	10.7 ± 1.9	3.8 ± 3.4	26.3 ± 0.6
FedTGP	25.3 ± 1.2	18.5 ± 1.3	26.1 ± 1.2	19.1 ± 1.0	9.9 ± 1.0	10.8 ± 1.7	4.5 ± 4.3	26.9 ± 0.6
FedGen	26.9 ± 2.0	19.7 ± 1.7	27.3 ± 1.7	20.5 ± 2.2	10.0 ± 1.2	8.6 ± 1.1	2.3 ± 0.3	22.4 ± 2.0
FedGH	31.9 ± 1.1	24.6 ± 1.0	35.7 ± 3.4	26.4 ± 1.3	12.8 ± 0.7	13.3 ± 0.8	3.5 ± 0.3	31.6 ± 1.7
LG-FedAvg	31.3 ± 0.8	24.4 ± 0.5	33.7 ± 1.1	25.6 ± 0.8	13.3 ± 0.4	13.3 ± 0.6	4.3 ± 0.5	29.7 ± 0.6
FedMRL	33.7 ± 1.0	26.7 ± 0.9	38.6 ± 3.7	28.3 ± 1.1	14.9 ± 1.0	14.9 ± 1.2	5.1 ± 1.0	32.7 ± 1.5

Table C.14: CIFAR-100-LT: main metrics (%), as Table C.13.

Method	Bal. acc.	Macro-F1	Precision	MCC	G-mean	Tail rec.	Worst rec.	AUC-PR
FedECS-LT	**18.7 ± 0.2**	**16.3 ± 0.3**	**19.2 ± 0.6**	**17.9 ± 0.2**	**10.5 ± 0.2**	**8.2 ± 0.2**	**0.2 ± 0.2**	**14.3 ± 0.3**
FedECS-LT (heads only)	14.3 ± 0.6	12.6 ± 0.5	15.7 ± 0.6	13.5 ± 0.6	7.9 ± 0.4	6.4 ± 0.3	0.2 ± 0.2	11.1 ± 0.5
GeFL-F	7.7 ± 0.4	4.7 ± 0.3	6.0 ± 0.4	6.9 ± 0.4	2.9 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	4.4 ± 0.2
Balanced Softmax	9.0 ± 0.3	5.8 ± 0.3	7.3 ± 0.4	8.2 ± 0.3	3.3 ± 0.1	2.2 ± 0.1	0.0 ± 0.0	5.1 ± 0.3
FedLC	8.5 ± 0.3	5.3 ± 0.2	7.7 ± 0.5	7.8 ± 0.3	3.0 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	5.0 ± 0.3
FedRS	8.5 ± 0.3	5.3 ± 0.2	7.9 ± 0.5	7.7 ± 0.3	2.9 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	5.0 ± 0.3
FedDF	10.0 ± 0.5	6.1 ± 0.3	9.7 ± 0.9	9.3 ± 0.5	3.3 ± 0.1	0.8 ± 0.1	0.0 ± 0.0	7.0 ± 0.4
Local	7.9 ± 0.8	4.9 ± 0.6	7.3 ± 0.7	7.2 ± 0.8	2.8 ± 0.2	1.6 ± 0.2	0.0 ± 0.0	4.8 ± 0.5
FedProto	4.8 ± 0.3	3.0 ± 0.1	3.9 ± 0.3	3.9 ± 0.3	2.2 ± 0.1	0.9 ± 0.0	0.0 ± 0.0	2.9 ± 0.1
FedTGP	4.5 ± 0.6	2.4 ± 0.4	3.4 ± 0.5	3.7 ± 0.6	1.9 ± 0.2	1.2 ± 0.5	0.0 ± 0.0	2.8 ± 0.6
FedGen	4.5 ± 1.3	2.6 ± 0.9	3.1 ± 0.8	3.7 ± 1.2	2.0 ± 0.3	0.7 ± 0.3	0.0 ± 0.0	2.2 ± 0.6
FedGH	6.5 ± 0.1	3.9 ± 0.2	5.0 ± 0.5	5.7 ± 0.1	2.5 ± 0.1	1.1 ± 0.1	0.0 ± 0.0	3.5 ± 0.1
LG-FedAvg	6.3 ± 0.1	3.7 ± 0.1	4.8 ± 0.2	5.4 ± 0.1	2.4 ± 0.0	1.0 ± 0.0	0.0 ± 0.0	3.3 ± 0.1
FedMRL	8.3 ± 0.2	5.0 ± 0.2	6.9 ± 0.1	7.5 ± 0.2	2.9 ± 0.1	1.7 ± 0.1	0.0 ± 0.0	4.4 ± 0.1

referee's decision boundaries, and its effect on accuracy is measured by the component ablation.

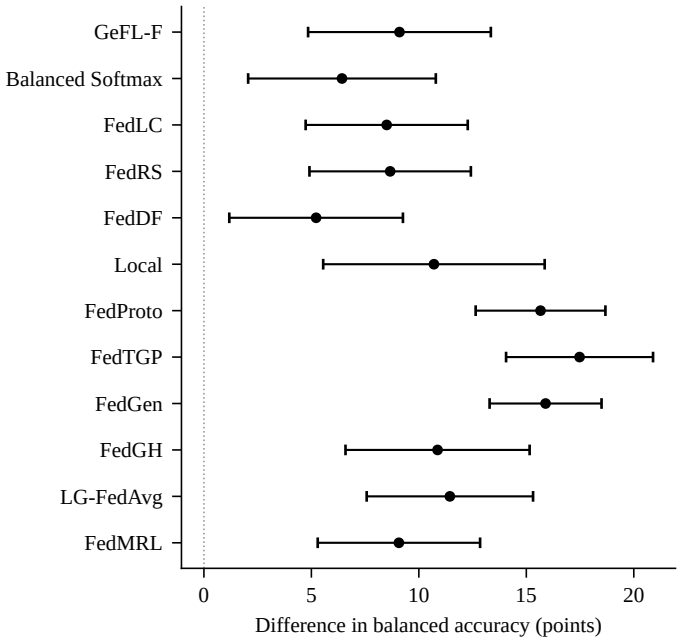


Figure C.5: Balanced accuracy of FedECS-LT minus that of each method on CIFAR-10-LT (points; heads as trained for every method, FedECS-LT included; mean of the paired differences over seeds 3–5 with 95% t confidence interval; right of the dotted line FedECS-LT is better). Imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$.

Table C.15: Generator fidelity (%): the referee's per-class accuracy on generated features, mean over the classes of each group (head: 30% most frequent, tail: 40% least frequent); mean ± standard deviation over seeds 3–5. FedDF uses GeFL-F's generator, trained in its own runs.

Dataset	Generator	All	Head	Medium	Tail
CIFAR-10	FedECS-LT (generator)	25.9 ± 2.1	11.0 ± 2.2	16.7 ± 13.0	44.1 ± 5.0
CIFAR-10	FedECS-LT (after moment correction)	28.9 ± 4.6	23.5 ± 6.6	19.2 ± 12.4	40.3 ± 2.8
CIFAR-10	GeFL-F (generator)	12.9 ± 2.4	9.4 ± 0.9	11.2 ± 7.6	16.8 ± 0.3
CIFAR-10	FedDF (generator)	12.8 ± 2.2	9.2 ± 1.2	10.3 ± 7.2	17.3 ± 0.7
CIFAR-100	FedECS-LT (generator)	8.8 ± 1.2	5.9 ± 1.8	8.2 ± 0.8	11.5 ± 1.2
CIFAR-100	FedECS-LT (after moment correction)	8.8 ± 0.4	7.2 ± 0.8	9.0 ± 0.8	9.9 ± 0.6
CIFAR-100	GeFL-F (generator)	1.1 ± 0.1	1.1 ± 0.2	1.1 ± 0.2	1.2 ± 0.2
CIFAR-100	FedDF (generator)	1.1 ± 0.1	1.2 ± 0.2	1.0 ± 0.2	1.1 ± 0.2

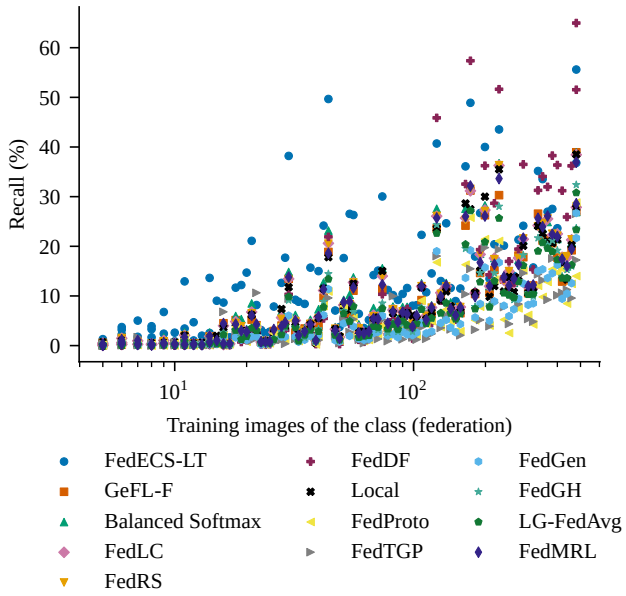


Figure C.6: Recall of every class against its number of training images in the federation (CIFAR-100-LT, imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$; heads as trained, before the readout of FedECS-LT; each point is one class, mean over seeds 3–5).

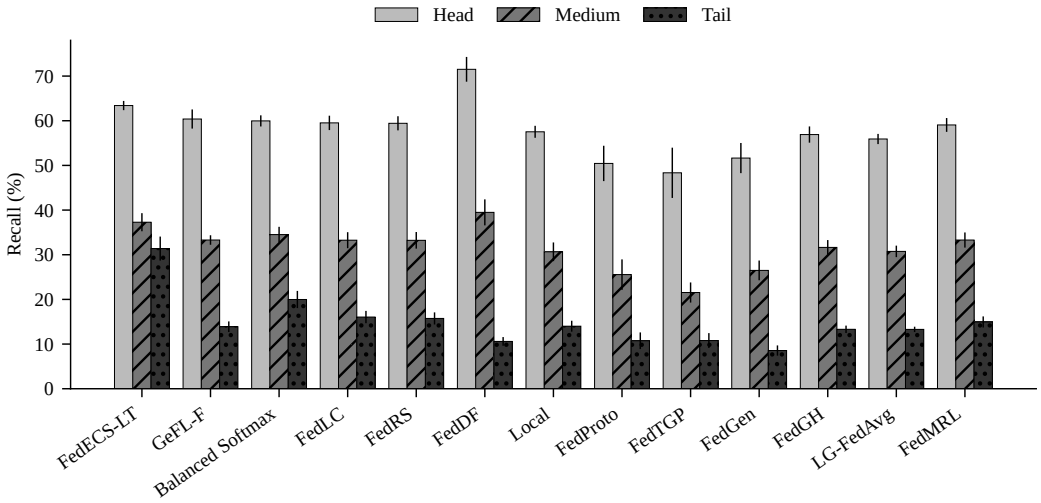


Figure C.7: Recall of the head (30% most frequent), medium (the next 30%) and tail (40% least frequent) classes, per method (CIFAR-10-LT, imbalance ratio 100, 10 clients, Dirichlet $\alpha = 0.5$; heads as trained, before the readout of FedECS-LT; mean $\pm$ standard deviation over seeds 3–5).

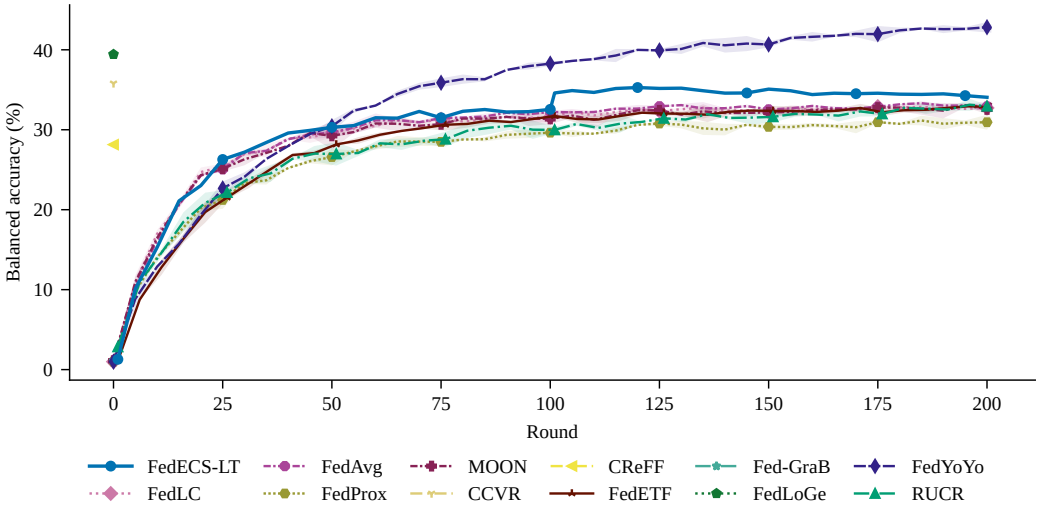


Figure C.8: Test balanced accuracy over the communication rounds with one shared ResNet-8 (CIFAR-100-LT, imbalance ratio 100, 20 clients, 40% per round, Dirichlet $\alpha = 0.5$; mean over seeds 3–5). For FedECS-LT the rounds are those of the extractor and then of the head.

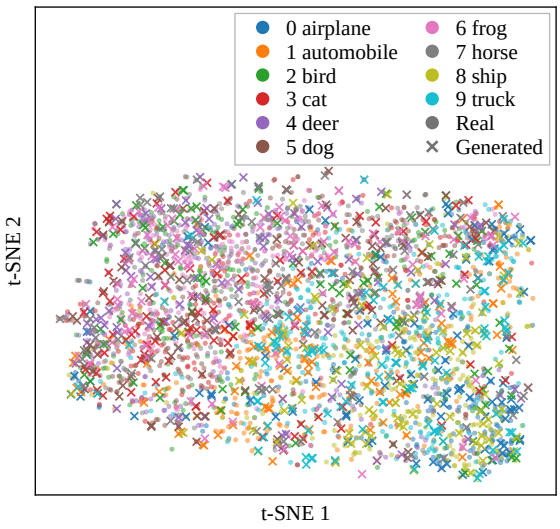


Figure C.9: Real validation features (circles) and features generated by FedECS-LT (crosses) on CIFAR-10-LT, embedded together by t-SNE (seed 3; same settings and caveats as Fig. 4).

Table C.16: GeFL's measure on the long-tailed benchmarks: each client model's best test accuracy (%) at GeFL's evaluation rounds of Stage III (the first, every tenth and the last), averaged over the ten models, as in GeFL's code; mean ± standard deviation over seeds 3–5. The main tables report the last round instead. For FedECS-LT only its heads: the readout is computed once, after the last round. FedDF: its client heads before the server's distillation step, since the per-round logs hold no distilled heads; its value can therefore be below its last-round value in Table 6.

Method	CIFAR-10	CIFAR-100
FedECS-LT (heads only)	46.1 ± 2.4	15.2 ± 0.5
GeFL-F	34.7 ± 1.3	8.4 ± 0.4
Balanced Softmax	**38.3 ± 1.8**	9.2 ± 0.4
FedLC	34.6 ± 1.3	8.6 ± 0.3
FedRS	34.4 ± 1.3	8.5 ± 0.3
FedDF	35.6 ± 1.2	**9.8 ± 0.5**
Local	32.1 ± 0.7	7.9 ± 0.8
FedProto	32.8 ± 1.3	7.6 ± 0.3
FedTGP	32.3 ± 1.4	7.6 ± 0.2
FedGen	32.1 ± 1.3	7.7 ± 0.2
FedGH	33.3 ± 1.4	7.8 ± 0.2
LG-FedAvg	33.0 ± 1.3	7.7 ± 0.2
FedMRL	33.9 ± 1.1	8.3 ± 0.2

Table C.17: Datasets with natural class imbalance, every method of the GeFL-F pipeline with the exact readout: balanced accuracy (%); mean ± standard deviation over seeds 3–5 (a value without ± is from one seed). Same setting as Table 10.

Method	DermaMNIST	OrganAMNIST	BloodMNIST	PathMNIST	EMNIST (ByClass)	FER-2013
FedECS-LT	**45.4 ± 1.9**	**83.3 ± 1.3**	86.3 ± 1.3	**73.6 ± 0.4**	**67.3**	**39.2 ± 1.0**
GeFL-F	44.6 ± 1.6	82.5 ± 1.0	86.3 ± 0.4	72.3 ± 0.5	66.5	38.2 ± 0.7
Balanced Softmax	42.8 ± 0.8	79.0 ± 1.4	82.9 ± 1.2	70.6 ± 0.6	64.5	35.7 ± 0.8
FedLC	43.0 ± 0.9	79.1 ± 1.3	83.0 ± 1.0	70.7 ± 0.7	64.6	35.6 ± 0.7
FedRS	43.1 ± 0.9	79.0 ± 1.3	82.9 ± 1.0	70.8 ± 0.8	64.5	35.5 ± 0.7
FedDF	44.7 ± 1.1	82.1 ± 1.3	**86.4 ± 0.5**	72.8 ± 0.1	66.6	37.7 ± 0.7

Table C.18: With the exact readout for every method of the GeFL-F pipeline: balanced accuracy of FedECS-LT minus each method (points), paired over seeds 3–5, with the 95% t interval; bold where significant after Holm's correction. Same setting as Table 8.

Method	CIFAR-10	CIFAR-100
GeFL-F	+2.3 [-0.4, +5.0]	**+3.8 [+3.1, +4.5]**
Balanced Softmax	+1.4 [-0.7, +3.6]	+1.5 [+0.6, +2.4]
FedLC	+1.4 [-0.6, +3.5]	+1.5 [+0.6, +2.4]
FedRS	+1.5 [-0.6, +3.5]	+1.5 [+0.6, +2.4]
FedDF	+1.6 [-1.1, +4.3]	**+2.9 [+2.1, +3.6]**

Table C.19: Recall of the tail classes (%) with the exact readout for every method of the GeFL-F pipeline; mean ± standard deviation over seeds 3–5. Same setting as Table 8.

Method	CIFAR-10	CIFAR-100
FedECS-LT	50.3 ± 1.5	8.2 ± 0.2
GeFL-F	47.5 ± 1.1	6.4 ± 0.4
Balanced Softmax	50.3 ± 0.3	**8.3 ± 0.4**
FedLC	**50.4 ± 0.2**	8.3 ± 0.4
FedRS	50.3 ± 0.2	8.3 ± 0.4
FedDF	48.1 ± 1.1	6.8 ± 0.3

Table C.20: Accuracy (%) on each client's own class mix (the test accuracy re-weighted to the class frequencies of the model's own clients) on the long-tailed benchmarks; FedECS-LT with its readout, its heads only, other methods without the readout; mean ± standard deviation over seeds 3–5.

Method	CIFAR-10	CIFAR-100
FedECS-LT	59.2 ± 1.8	28.3 ± 0.3
FedECS-LT (heads only)	68.6 ± 1.9	31.5 ± 0.3
GeFL-F	73.0 ± 2.3	31.1 ± 0.6
Balanced Softmax	72.8 ± 2.3	33.1 ± 0.7
FedLC	74.2 ± 2.1	34.7 ± 0.7
FedRS	**74.2 ± 2.1**	**34.8 ± 0.8**
FedDF	71.5 ± 2.2	32.9 ± 1.1
Local	72.7 ± 2.6	32.3 ± 2.4
FedProto	61.3 ± 1.1	17.0 ± 1.3
FedTGP	62.7 ± 3.4	21.1 ± 1.3
FedGen	67.0 ± 2.7	22.0 ± 4.2
FedGH	72.6 ± 2.3	29.4 ± 0.8
LG-FedAvg	71.0 ± 2.2	28.0 ± 0.4
FedMRL	73.5 ± 2.2	33.7 ± 0.7

Table C.21: Calibration on the long-tailed benchmarks: expected calibration error (%, 15 bins; lower is better); FedECS-LT with its readout, its heads only, other methods without the readout; mean ± standard deviation over seeds 3–5.

Method	CIFAR-10	CIFAR-100
FedECS-LT	21.2 ± 0.5	43.8 ± 0.4
FedECS-LT (heads only)	37.8 ± 4.0	41.4 ± 0.9
GeFL-F	48.0 ± 2.2	42.4 ± 0.7
Balanced Softmax	55.2 ± 0.9	74.3 ± 0.4
FedLC	58.1 ± 0.8	75.8 ± 0.4
FedRS	58.3 ± 0.8	76.0 ± 0.5
FedDF	35.9 ± 1.9	31.8 ± 1.2
Local	53.2 ± 3.8	69.0 ± 3.4
FedProto	17.1 ± 0.8	**3.8 ± 0.3**
FedTGP	**8.8 ± 2.9**	25.8 ± 13.3
FedGen	67.3 ± 1.7	73.9 ± 13.9
FedGH	57.3 ± 0.8	71.2 ± 0.3
LG-FedAvg	61.0 ± 0.3	77.2 ± 0.2
FedMRL	60.2 ± 0.9	77.5 ± 0.5

Table C.22: Fairness across clients on the long-tailed benchmarks: balanced accuracy (%) of the weakest head architecture; FedECS-LT with its readout, its heads only, other methods without the readout; mean ± standard deviation over seeds 3–5.

Method	CIFAR-10	CIFAR-100
FedECS-LT	**48.5 ± 1.5**	**14.9 ± 1.9**
FedECS-LT (heads only)	38.8 ± 1.9	11.7 ± 0.9
GeFL-F	27.5 ± 2.4	6.0 ± 0.8
Balanced Softmax	30.4 ± 0.9	7.1 ± 1.1
FedLC	28.1 ± 1.1	6.8 ± 0.9
FedRS	27.8 ± 1.1	6.8 ± 0.8
FedDF	32.2 ± 2.8	8.4 ± 1.5
Local	25.5 ± 2.6	5.1 ± 1.5
FedProto	18.7 ± 3.3	3.5 ± 0.2
FedTGP	16.8 ± 1.9	1.7 ± 0.7
FedGen	20.5 ± 1.6	3.1 ± 1.8
FedGH	25.6 ± 2.2	5.2 ± 0.5
LG-FedAvg	25.3 ± 1.1	4.9 ± 0.5
FedMRL	27.4 ± 0.9	7.1 ± 0.3